

MACHINE LEARNING-DRIVEN FAULT DETECTION IN SWITCHED RELUCTANCE MOTOR

Project Report submitted

In Partial Fulfilment of the Requirements for the award of the degree

of

Bachelor of Technology

In

Electrical and Electronics Engineering

by

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**DEPARTMENT OF
ELECTRICAL AND ELECTRONICS ENGINEERING**

**VELAGAPUDI RAMAKRISHNA
SIDDHARTHA ENGINEERING COLLEGE (AUTONOMOUS)**
*(Affiliated to JNTUK, Approved by UGC, Accredited By NBA and
NAAC A+ Grade)*

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DEPARTMENT OF ELECTRICAL AND ELECTRONICS ENGINEERING

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CERTIFICATE

This is to certify that the MAJOR PROJECT report titled “**MACHINE LEARNING-DRIVEN FAULT DETECTION IN SWITCHED RELUCTANCE MOTORS**” is a bonafide record of work done by **A. KOWSHIK (218W1A0201), K. RAVI RAJA (218W1A0214), D. RAVI SUNDAR (218W1A0208), P. SNIGDHA (218W1A0233)** under my guidance and supervision and is submitted in partial fulfilment of the requirements for the award of the degree of Bachelor of Technology in Electrical & Electronics Engineering, V.R. Siddhartha Engineering College (Autonomous) during the academic year 2024-2025.

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DECLARATION

We hereby declare that the work is being presented in this MAJOR PROJECT report “**MACHINE LEARNING -DRIVEN FAULT DETECTION IN SWITCHED RELUCTANCE MOTORS**” submitted towards the partial fulfillment of requirements for the award of the degree of **Bachelor of Technology in Electrical and Electronics Engineering** in V. R. Siddhartha Engineering College, Vijayawada is an authentic record of our work carried out under the supervision of **Dr.A.Veera Reddy, Assistant professor**, in EEE Department, in V. R. Siddhartha Engineering College, Vijayawada. The matter embodied in this dissertation report has not been submitted by us for the award of any other degree. Furthermore, the technical details furnished in various chapters of this report are purely relevant to the above **MAJOR PROJECT**.

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ABSTRACT

The evolution of artificial intelligence has simplified human effort in nearly every aspect of electrical and electronic appliance operations, as well as in their maintenance. This project focuses to detect faults in Switched Reluctance Motor (SRM) by applying the machine learning approach through analysis of relevant motor parameters such as current, flux, torque, and induced voltage in the system using Finite Element Method (FEM). These motor parameters serve as crucial indicators of the SRM's health and can provide valuable insights into potential faults. Additionally, these parameters help in creating a comprehensive dataset for training the machine learning model. Although the Switched Reluctance Motor is among the most fault-tolerant motors, it is not entirely resistant to faults. Therefore, it is essential to monitor the motor's operation to effectively detect faults and ensure proper maintenance. This can be achieved by employing random forest method for faster and more reliable fault detection compared to manual monitoring. The dataset derived from the simulation of the motor at different conditions using FEM. A high degree of accuracy can be achieved by developing a robust algorithm based on this approach. This will ensure more stable operation of the motor under adverse working conditions. Additionally, the use of predictive maintenance strategies will help reduce downtime and increase the lifespan of the SRM. By continuously monitoring motor performance, the system can also optimize energy efficiency, contributing to more sustainable and cost-effective operation. Furthermore, the implementation of this system can be extended to real-time applications for dynamic fault detection, ensuring timely interventions. The integration of AI-driven monitoring tools will lead to a more intelligent and autonomous maintenance system for SRMs.

TABLE OF CONTENTS

TITLE	PAGE NO.
DECLARATION	iii
ACKNOWLEDGEMENT	iv
ABSTRACT	v
TABLE OF CONTENTS	vi
LIST OF FIGURES	vii
CHAPTER 1	
INTRODUCTION	
1.1 Motivation	1
1.2 Literature survey	2
1.3 Proposed system Description	3
CHAPTER 2	
SWITCHED RELUCTANCE MOTOR AND AIRGAP FAULTS IN SRM	
2.1 Introduction	4
2.2 Faults in SRM	5
2.3 Types of Eccentricity	6
2.3.1 Static Eccentricity	7
2.3.2 Dynamic Eccentricity	8
2.3.3 Mixed Eccentricity	9
CHAPTER 3	
DESIGN OF SWITCHED RELUCTANCE MOTOR	
3.1 RMXprt Design of SRM	10
3.2 2D Design of SRM	17
3.3 Fault generation in SRM	19

CHAPTER 4

MACHINE LEARNING TECHNIQUES

4.1	Machine learning approach	20
4.1.1	Linear regression	20
4.1.2	Random Forest	21
4.1.3	Support Vector Machine (SVM)	22
4.1.4	Artificial Neural Network	22
4.1.5	Selection of Best Algorithm for the project	22
4.2	Algorithm implementation	23
4.2.1	Python for Machine Learning	24
4.2.2	Jupyter Notebook for Model development	25
4.2.3	Libraries used	26

CHAPTER 5

ML MODEL FOR FAULT ANALYSIS IN SRM

5.1	Importing required libraries	28
5.2	Loading and processing the datasets	29
5.3	Defining features and Target variable	29
5.4	Splitting the Dataset into Training and Testing sets	30
5.5	Training the Machine learning model	31
5.6	Evaluating the Model's Accuracy	31
5.7	Creating a function for Real-time Fault Prediction	31
5.8	Running the Fault Prediction Function	31

CHAPTER 6

RESULTS & DISCUSSION

6.1	Simulation Results	32
6.2	Machine Learning Algorithm Results	38

CHAPTER 7

CONCLUSION AND FUTURE SCOPE	41
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REFERENCES	42
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LIST OF FIGURES

S.NO	FIGURE	PAGE NO
1	Switched Reluctance Motor	4
2	Static Eccentricity	7
3	Dynamic Eccentricity	8
4	Mixed Eccentricity	9
5	Maxwell Ansoft software	10
6	Selecting Switched Reluctance Motor	10
7	Creation of RMXprt model	11
8	Machine parameters of SRM	12
9	Circuit parameters of SRM	12
10	Stator Parameters of SRM	13
11	Winding parameter of SRM	13
12	Rotor parameters of SRM	14
13	Shaft parameters of SRM	14
14	Setup configuration	15
15	Design model of SRM	15
16	Performance of SRM	16
17	Creating Maxwell 2D design	17
18	2D design of SRM	18
19	Selecting rotor in SRM	19
20	Scale of the rotor	19
21	Linear regression	20
22	Random forest	21
23	Efficiency graph of SRM	32
24	Flux linkage vs current at various positions	32
25	Torque graph at normal condition	33
26	Winding current at normal condition	33
27	Torque graph at fault-1 condition	34
28	Winding current at fault-1 condition	34
29	Torque graph at fault-2 condition	35
30	Winding current at fault-2 condition	35

31	Torque graph at fault-3 condition	36
32	Winding current at fault-3 condition	36
33	Torque graph at fault-4 condition	37
34	Winding current at fault-4 condition	37
35	Machine learning output at normal state	38
36	Machine learning output at fault in phase A	38
37	Machine learning output at fault in phase B	39
38	Machine learning output at fault in phase C	40
39	Machine learning output at fault in phase D	40

CHAPTER 1

INTRODUCTION

Artificial intelligence (AI) has made things easier by improving the way we operate and maintain electrical appliances. One area where AI can be especially helpful is in detecting faults in Switched Reluctance Motors (SRMs). These motors are known for being quite reliable, but they can still develop problems over time. Detecting these problems early is important to avoid costly downtime and ensure smooth operation.

This project focuses on using machine learning to identify faults in SRMs by looking at key motor parameters, such as current, flux, torque, and voltage, which can tell us how healthy the motor is. These parameters are gathered through simulations using a method called Finite Element Method (FEM), which helps create a dataset for training a machine learning model.

To make fault detection faster and more accurate, we use the Random Forest method, a popular machine learning technique. This system can monitor the motor's performance continuously, reducing the need for manual checks. By detecting issues early, we can keep the motor running smoothly for longer, saving energy and cutting costs.

The project also aims to make this system work in real-time, so it can quickly identify problems and take action as needed. With AI-driven monitoring, SRMs can be maintained in a smarter, more automated way, helping them perform better and last longer.

1.1 Motivation

This project focuses on enhancing the reliability and performance of Switched Reluctance Motors (SRMs) by leveraging machine learning for fault detection. SRMs are known for their robustness and fault tolerance, yet they are not immune to failures, which can disrupt operations and lead to costly downtime. Traditionally, fault detection in SRMs relies on manual inspection, which can be time-consuming and may not always identify issues before they escalate.

By utilizing machine learning, this project aims to monitor key motor parameters such as current, flux, torque, and induced voltage that are critical indicators of the motor's health. These parameters are obtained through simulations using the Finite Element Method (FEM),

which helps generate a comprehensive dataset for training the machine learning model.

The Random Forest method is applied to improve the accuracy and speed of fault detection, providing a more reliable alternative to manual monitoring. The proposed system offers continuous, real-time monitoring of motor performance, enabling early fault detection and reducing the risk of unexpected failures. This predictive maintenance approach not only ensures stable motor operation but also extends the lifespan of SRMs by preventing unnecessary wear and tear. In addition, by optimizing energy efficiency, the system contributes to more cost-effective and sustainable operations.

1.2 Literature Survey

Switched Reluctance Motors (SRMs) are widely recognized for their robust design, simple construction, and high fault tolerance. Despite these advantages, SRMs are still susceptible to various mechanical and electrical faults that can affect their performance. To address these challenges, recent research has focused on integrating advanced simulation tools and machine learning techniques for effective fault detection and predictive maintenance.

Finite Element Method (FEM) plays a critical role in accurately modeling the electromagnetic behavior of SRMs. It enables detailed analysis of motor parameters such as current, torque, flux, and induced voltage under normal and faulty conditions. This simulation-based data is essential for training machine learning models, as it provides a comprehensive understanding of how faults affect motor performance.

Traditional fault detection methods relied on physical inspections or basic signal analysis, which often lacked precision and required significant manual effort. Modern approaches utilize machine learning algorithms, such as Random Forest, to automatically detect and classify faults based on patterns observed in motor parameter data. These data-driven methods offer faster and more reliable diagnostics.

There has also been a shift towards using condition monitoring systems that continuously track motor health and detect anomalies before failure occurs. This allows for predictive maintenance, reducing unplanned downtimes and extending the operational life of the motor. Combining machine learning with FEM simulation enhances the accuracy of fault detection and contributes to energy efficiency by optimizing motor operation.

Overall, the integration of AI, especially machine learning, with FEM-based modeling represents a powerful approach to SRM fault analysis. This not only improves diagnostic precision but also enables the implementation of real-time monitoring systems suitable for industrial applications. The advancements support a shift from reactive to proactive maintenance strategies, making motor systems more intelligent, efficient, and reliable.

1.3 Proposed system description

1. Simulation of SRM Using FEM:

The first step in the proposed system involves simulating the Switched Reluctance Motor (SRM) using the Finite Element Method (FEM). This allows us to model the motor's behavior in both normal and fault conditions. By doing this, we can monitor key parameters like current, flux, torque, and induced voltage-critical indicators of the motor's health. The simulation generates a dataset that reflects how the motor performs in different scenarios, laying the groundwork for training the machine learning model.

2. Data Collection and Preprocessing:

Next, the data from the simulations is collected and processed. This involves organizing the data, normalizing it, and selecting the most relevant features that help in fault detection.

Labels are also applied to the data to indicate whether the motor is in a normal or faulty condition. This prepares the data for the machine learning model, ensuring that the model can accurately learn the patterns associated with different motor states.

3. Development of Machine Learning Model:

With the cleaned and labelled dataset in hand, a machine learning model is trained using the Random Forest algorithm. This model learns to recognize patterns in the motor's parameters that signal potential faults. By training and testing the model with real data, we ensure that it can predict faults accurately and quickly.

4. Fault Prediction and Monitoring:

Once the model is trained, it is integrated into a real-time monitoring system. This system continuously collects motor data and uses the trained model to predict faults before they occur. If a potential fault is detected, the system alerts the maintenance team so they can intervene before the issue escalates. This proactive approach minimizes downtime.

CHAPTER 2

SWITCHED RELUCTANCE MOTOR AND AIRGAP FAULTS IN SRM

2.1 Introduction

The Switched Reluctance Motor (SRM) operates on the principle of magnetic reluctance, where the rotor seeks the path of least magnetic resistance. Unlike traditional motors, SRMs don't use permanent magnets or rotor windings. Instead, they feature salient poles on both the stator and rotor. The working principle involves sequentially energizing the stator windings, creating a rotating magnetic field that pulls the rotor poles into alignment. This continuous switching of stator currents, controlled by electronic switches, generates the rotational motion as shown in Fig.2.1.

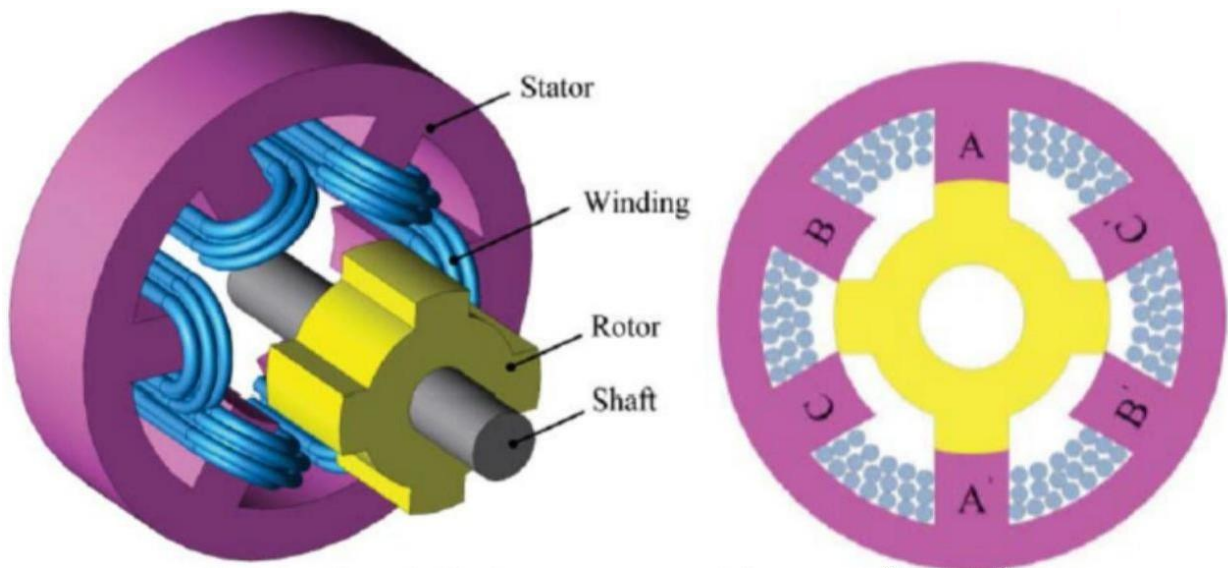


Fig.2.1 Switched Reluctance Motor

The Switched Reluctance Motor (SRM) functions based on the principle of variable magnetic reluctance. Both the stator and rotor have salient poles, meaning they have projecting teeth. When a specific stator winding is energized, it creates a magnetic field. This field induces a magnetic force that pulls the nearest rotor pole into alignment with the energized stator pole, minimizing the reluctance (resistance to magnetic flux). As the rotor aligns, the energized winding is switched off, and the next stator winding in sequence is energized. This continuous, sequential energizing of stator windings creates a rotating magnetic field, causing the rotor to rotate. The speed and torque of the motor are controlled by adjusting the timing and magnitude of the current pulses applied to the stator windings.

Advantages:

- **Robustness and Durability:** The simple construction, lacking permanent magnets and rotor windings, makes SRMs highly resilient to harsh environments.
- **High Efficiency:** Their design minimizes losses, contributing to excellent efficiency.
- **Wide Speed Range and Torque Control:** SRMs offer precise control over speed and torque, making them suitable for variable-speed applications.
- **Cost-Effectiveness:** The absence of expensive rare-earth magnets reduces production costs.
- **High Fault Tolerance:** Because the phase windings are independent, if one phase fails, the motor can often continue to operate.

Applications:

- **Electric Vehicles:** SRMs are gaining traction in electric and hybrid vehicles due to their high torque density and efficiency.
- **Industrial Automation:** Their precise control makes them ideal for robotics, machine tools, and conveyor systems.
- **Appliances and HVAC Systems:** SRMs enhance the performance and energy efficiency of devices like washing machines and vacuum cleaners.
- **Renewable Energy Systems:** They are used in wind turbines and solar trackers for efficient energy conversion.

2.2 Faults in Switched Reluctance Motor

Switched Reluctance Motors (SRMs), while inherently robust due to their simple construction, are not immune to faults. The operational demands placed on SRMs, particularly in high-stress applications like electric vehicles and industrial automation, can lead to a variety of electrical and mechanical failures. These faults, if undetected, can significantly impact the motor's performance, efficiency, and longevity, potentially leading to catastrophic failures. The occurrence of faults in SRMs stems from a combination of factors, including the harsh operating environments, the complex electronic control required for their operation, and the inherent stresses associated with their pulsed torque generation.

- **Converter Faults:** Faults related to the power converter that supplies the SRM, such as DC voltage fluctuations or failures in the control circuitry.
- **Bearing Faults:** Wear and tear, lubrication issues, or misalignment can lead to bearing failures. This can cause vibrations, noise, and eventually, rotor damage.
- **Rotor Eccentricity:** This occurs when the rotor is not centered within the stator, resulting in an uneven air gap. This can lead to unbalanced magnetic forces and increased vibrations.
- **Shaft Misalignment:** Improper alignment of the motor shaft with the load can cause excessive stress on the bearings and other mechanical components.
- **Sensor Faults:** Involve errors or failures in position or speed sensors, affecting accurate commutation and control of the motor.
- **Open Circuit Fault:** Happens when there is a break in the winding or driver circuit, leading to the affected phase failing to produce torque
- **Phase-to-Phase Short Circuit:** Fault between two different phase windings that causes high current flow, imbalance, and may damage the motor drive or power electronics.
- **Combined Eccentricity:** A combination of static and dynamic eccentricities, resulting in more complex and severe distortion of magnetic fields and torque production.

2.3 Types of Eccentricity

Eccentricity in a Switched Reluctance Motor (SRM) refers to the condition where the rotor's rotational axis is not perfectly aligned with the stator's center. This misalignment creates an uneven air gap between the rotor and stator poles, resulting in unbalanced magnetic forces during motor operation. This uneven air gap distribution can lead to increased vibrations, noise, and potentially, premature bearing wear. Furthermore, eccentricity can affect the motor's performance by reducing torque output and efficiency, and in severe cases, it can cause rotor-stator contact, leading to catastrophic failure.

2.3.1 Static Eccentricity (SE):

In static eccentricity, the rotor is misaligned and fixed in an off-center position within the stator. This creates an uneven air gap, causing an unbalanced magnetic pull (UMP) that leads to localized flux concentration as shown in Fig.2.2. As a result, certain stator regions experience higher saturation, increasing losses, vibrations, and mechanical stress. The motor's efficiency drops, and excessive heating can occur in specific areas. If left uncorrected, it may lead to premature wear of bearings and other components.

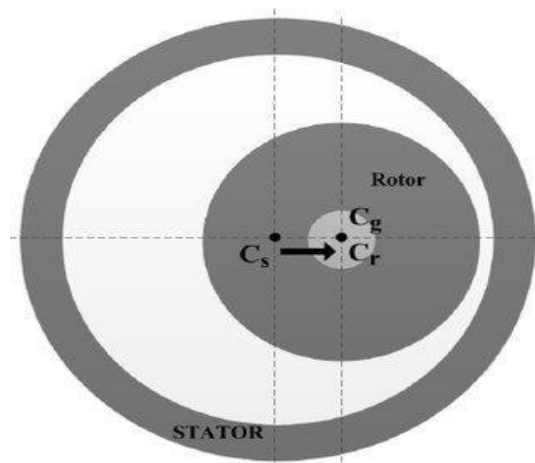


Fig.2.2 Static Eccentricity

Cause:

- Manufacturing imperfections leading to off-center rotor mounting
- Improper assembly or alignment of the rotor and stator
- Shaft bending or deformation
- Bearing wear or misplacement during motor installation

Effect:

- Uneven airgap between stator and rotor
- Asymmetric magnetic field distribution
- Increased vibration and acoustic noise
- Rise in unbalanced magnetic pull, leading to mechanical stress
- Potential degradation of motor efficiency and performance over time

2.3.2 Dynamic Eccentricity (DE):

In dynamic eccentricity, the rotor is misaligned and its position changes as it rotates, causing a varying air gap. This leads to fluctuating magnetic forces, creating periodic unbalanced magnetic pull (UMP) that results in torque ripple and vibrations as shown in Fig.2.3. The continuous variation in air gap flux can increase core losses and electromagnetic noise. Over time, it may accelerate bearing wear and mechanical instability. If not addressed, it can lead to performance degradation and potential motor failure

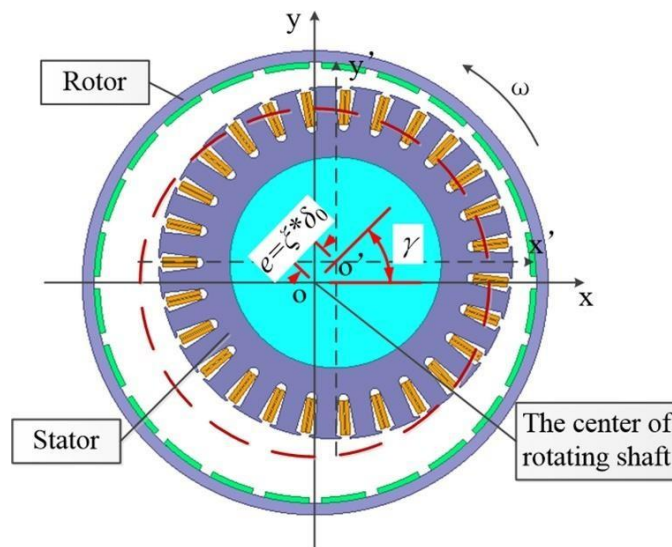


Fig.2.3 Dynamic Eccentricity

Cause:

- Rotor shaft misalignment during rotation
- Bent or unbalanced rotor shaft
- Worn or damaged bearings
- Inadequate support or looseness in the rotor system
- Vibration-induced displacement during motor operation

Effect:

- Time-varying airgap during rotor rotation
- Fluctuating magnetic field and unsteady torque output
- Increased torque ripple and noise
- Potential interaction with other mechanical faults (e.g., bearing wear)

2.3.3 Mixed Eccentricity (ME):

Mixed eccentricity is a combination of both static eccentricity (SE) and dynamic eccentricity (DE), meaning the rotor is misaligned and fixed off-center while also shifting its position dynamically as it rotates as shown in Fig.2.4. This results in an irregular and constantly changing air gap, causing complex variations in magnetic forces and flux distribution.

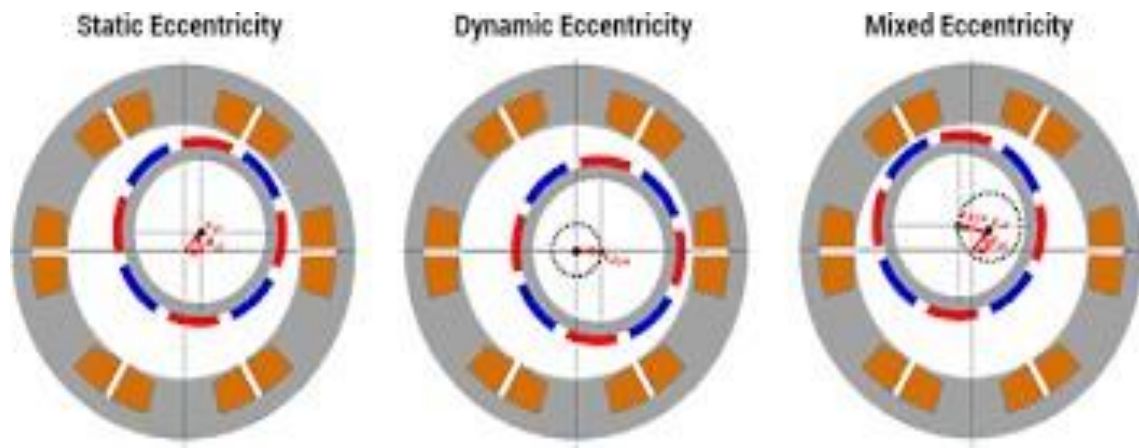


Fig.2.4 Mixed Eccentricity

Cause:

- Simultaneous presence of both static and dynamic eccentricity
- Progressive bearing degradation leading to both fixed and rotating shaft offset
- Structural imbalance combined with rotor misalignment
- Long-term mechanical stress or wear affecting both motor mounting and rotor dynamics

Effect:

- Highly irregular and asymmetrical airgap variation during rotation
- Severe distortion of magnetic flux distribution
- Excessive torque ripple, acoustic noise, and vibration
- Accelerated mechanical wear and potential for early motor failure
- Complicated fault detection and diagnosis due to overlapping fault signatures

CHAPTER 3

DESIGN OF SWITCHED RELUCTANCE MOTOR

3.1 RMXprt Design

Maxwell Ansoft is preferred for simulations due to its accurate Finite Element Analysis (FEA) for electromagnetic modeling. The software integrates with Ansys tools for Multiphysics analysis, including thermal and structural effects.

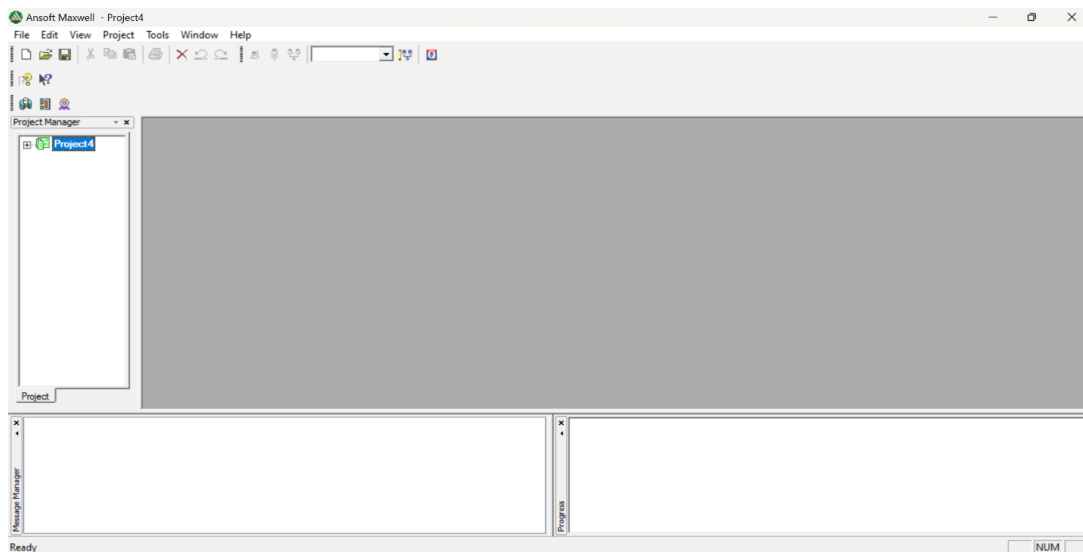


Fig.3.1 Maxwell Ansoft Software

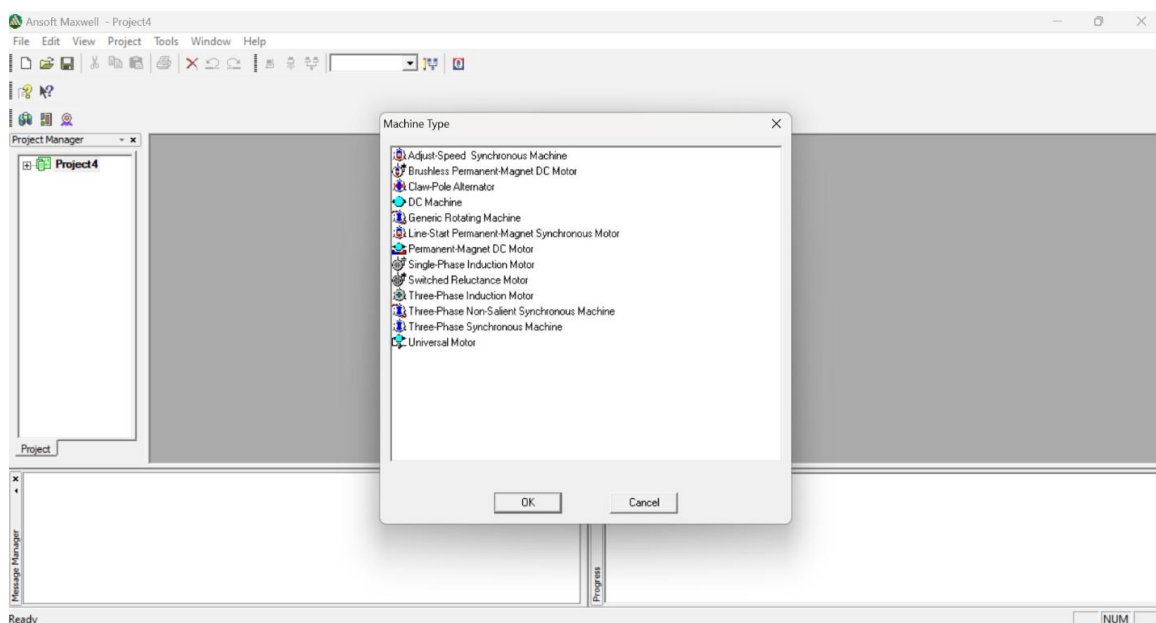


Fig.3.2 Selecting Switched Reluctance motor

The project is titled with specific name as required. so in order to design the motor, selection of motor is to be done. Click on project in menu and select Insert RMXprt design which shows various types of motor as shown in Fig.3.1 and Fig.3.2. Select the required motor for designing as the Switched Reluctance motor is Required select the motor.

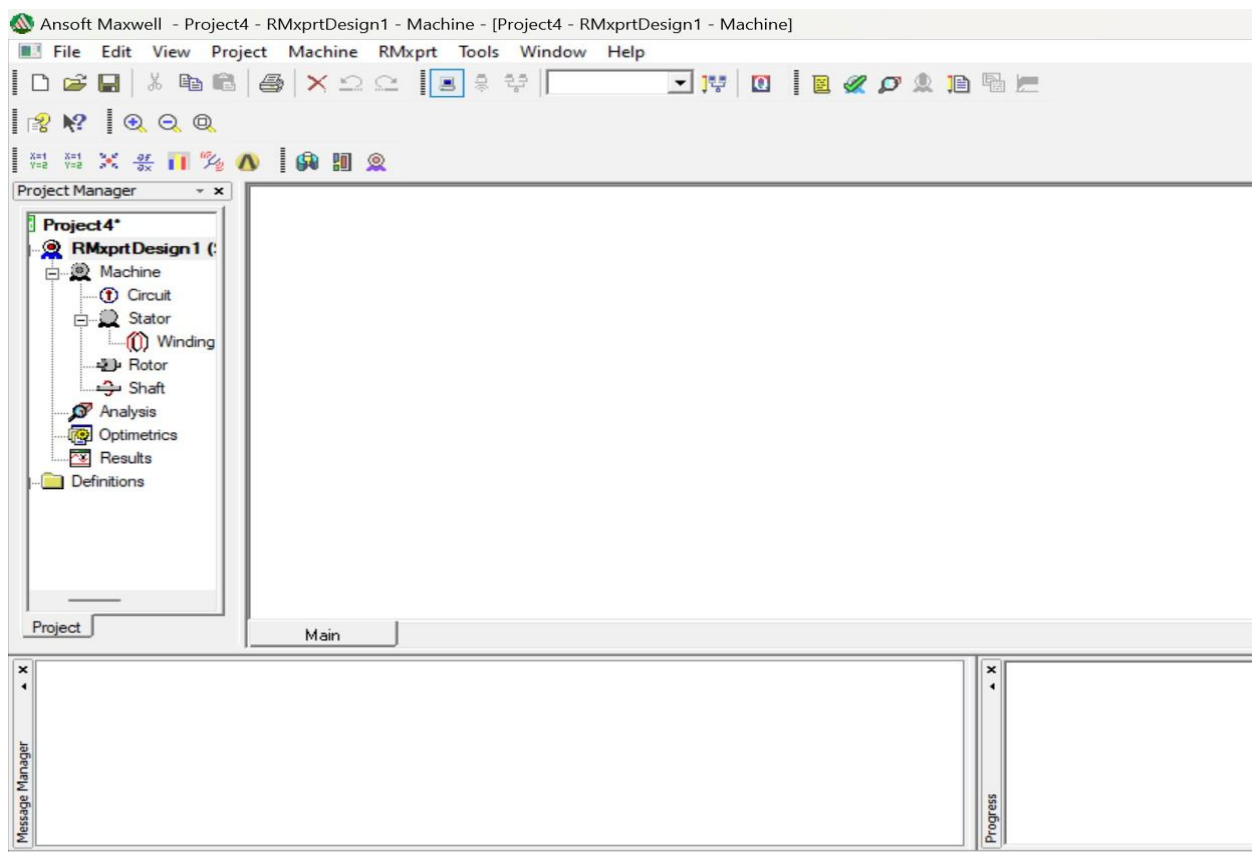


Fig.3.3 Creation of RMXPRT model

The Project is created to design switched Reluctance motor. The machine consists of its Stator, Circuit, Winding, Rotor, Shaft. The parameters are to be initialized each based on the required motor specifications, the values must present in between the range of standard SRM for better and real time performance analysis. The parameters are initialized as shown in Fig.3.3 - 3.10.

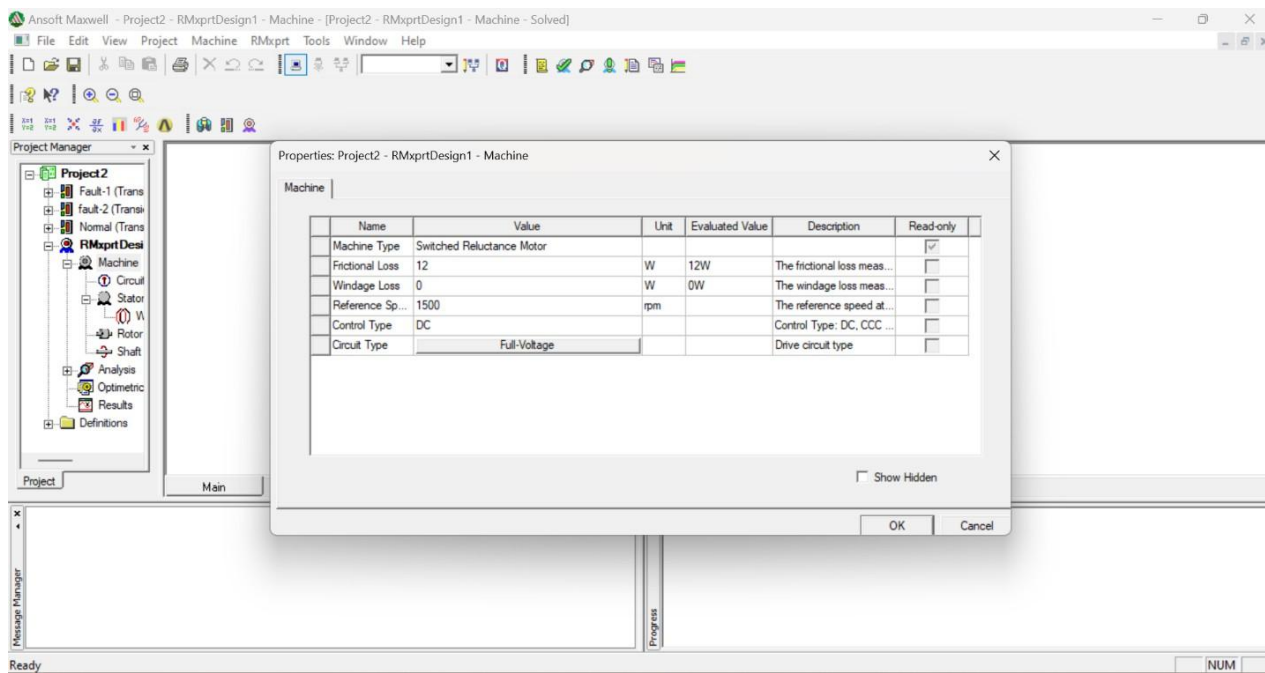


Fig.3.4 Machine Parameters of SRM

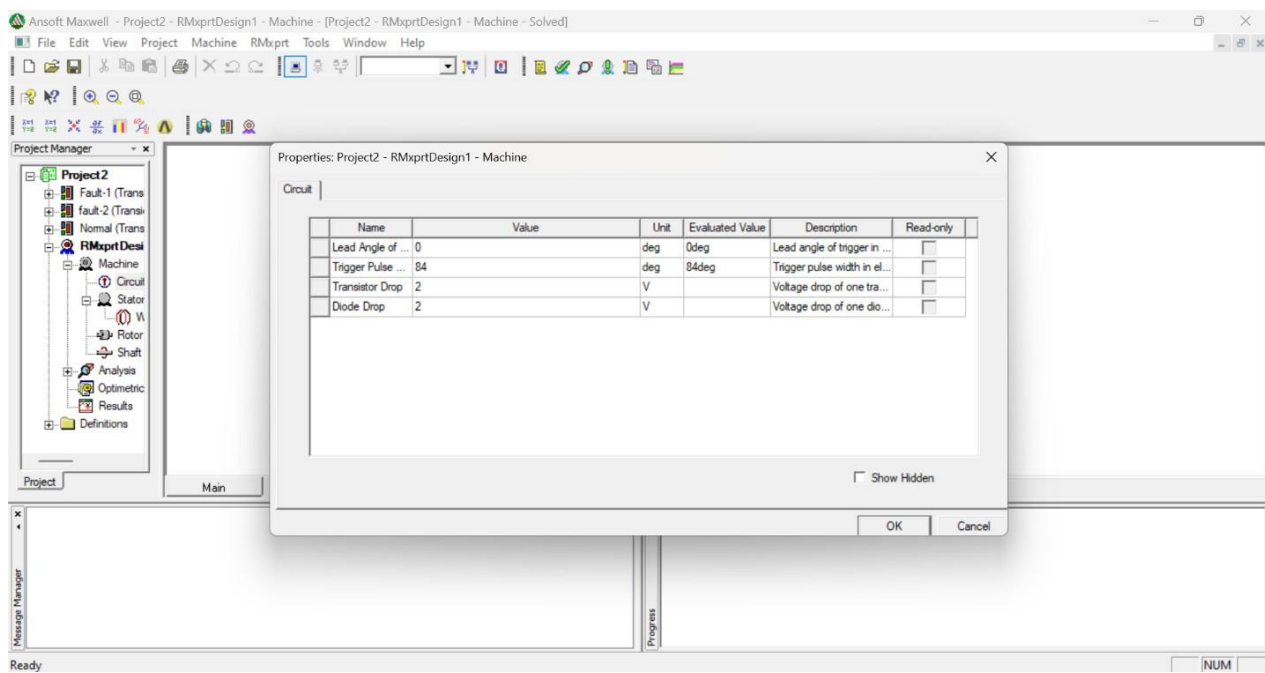


Fig.3.5 Circuit Parameters OF SRM

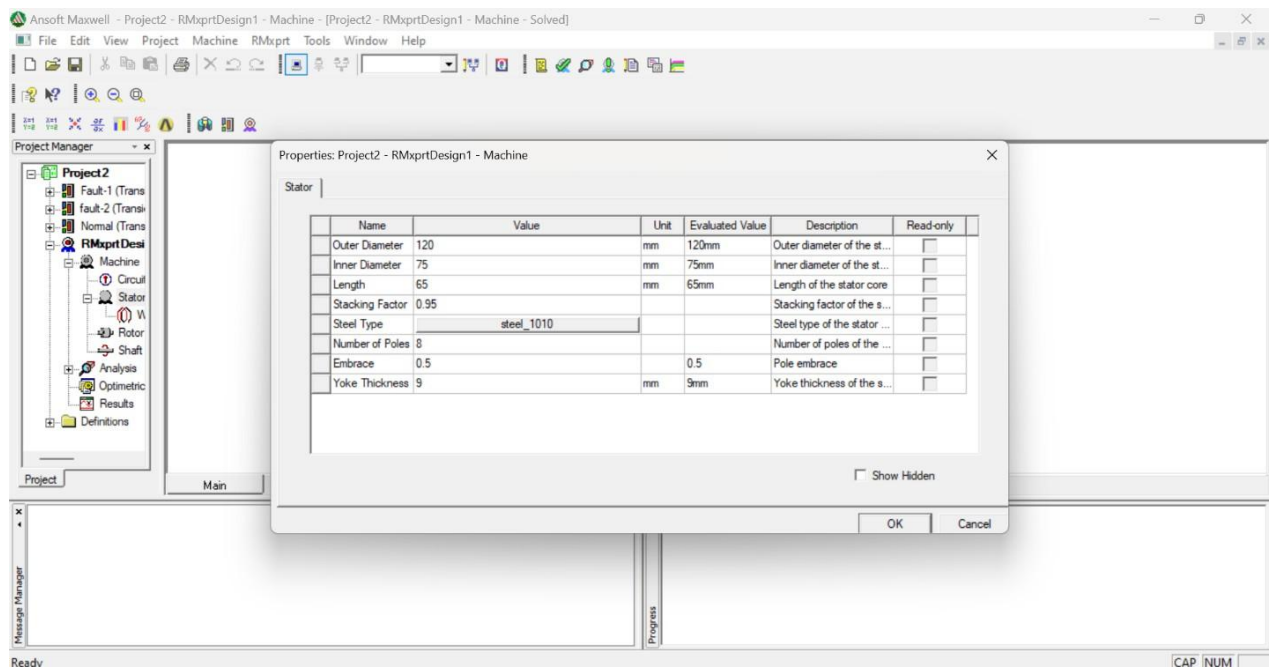


Fig.3.6 Stator Parameters of SRM

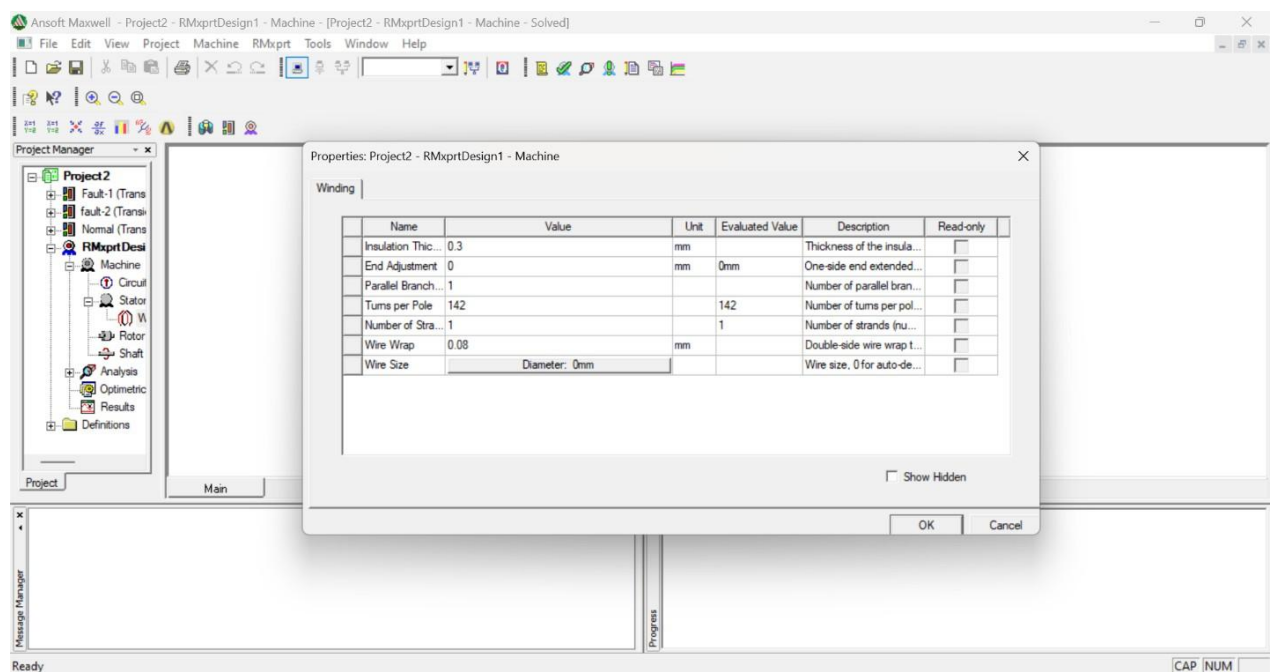


Fig.3.7 Winding Parameters of SRM

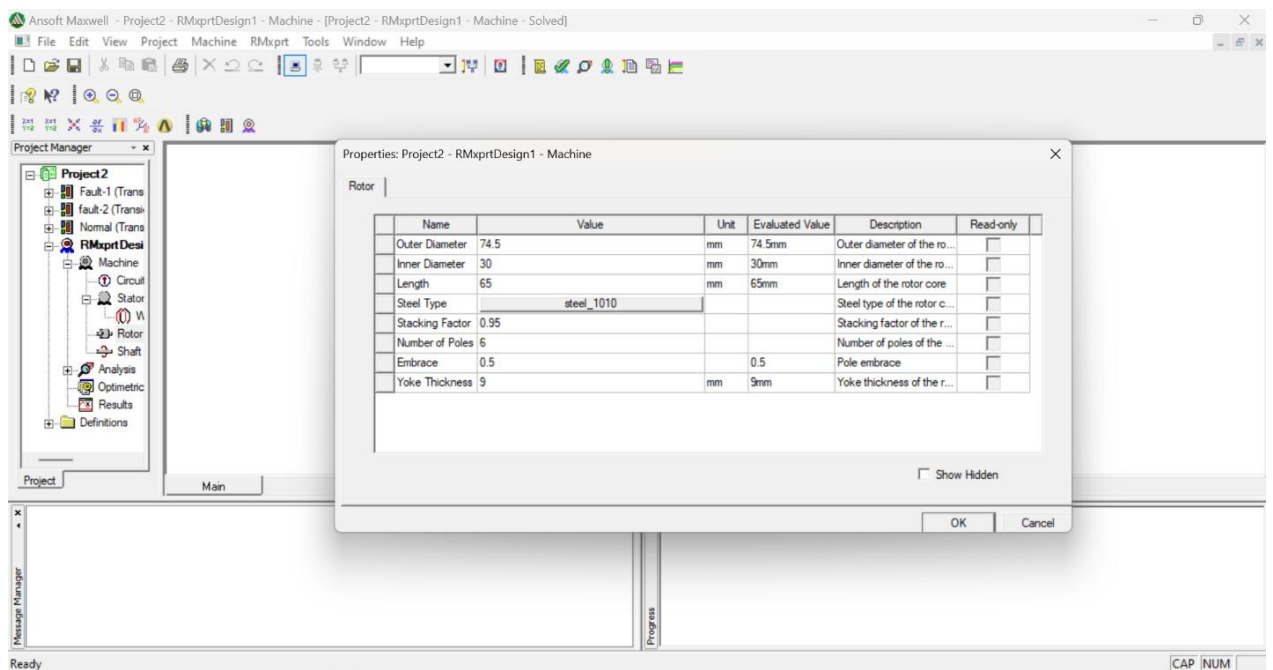


Fig.3.8 Rotor Parameters of SRM

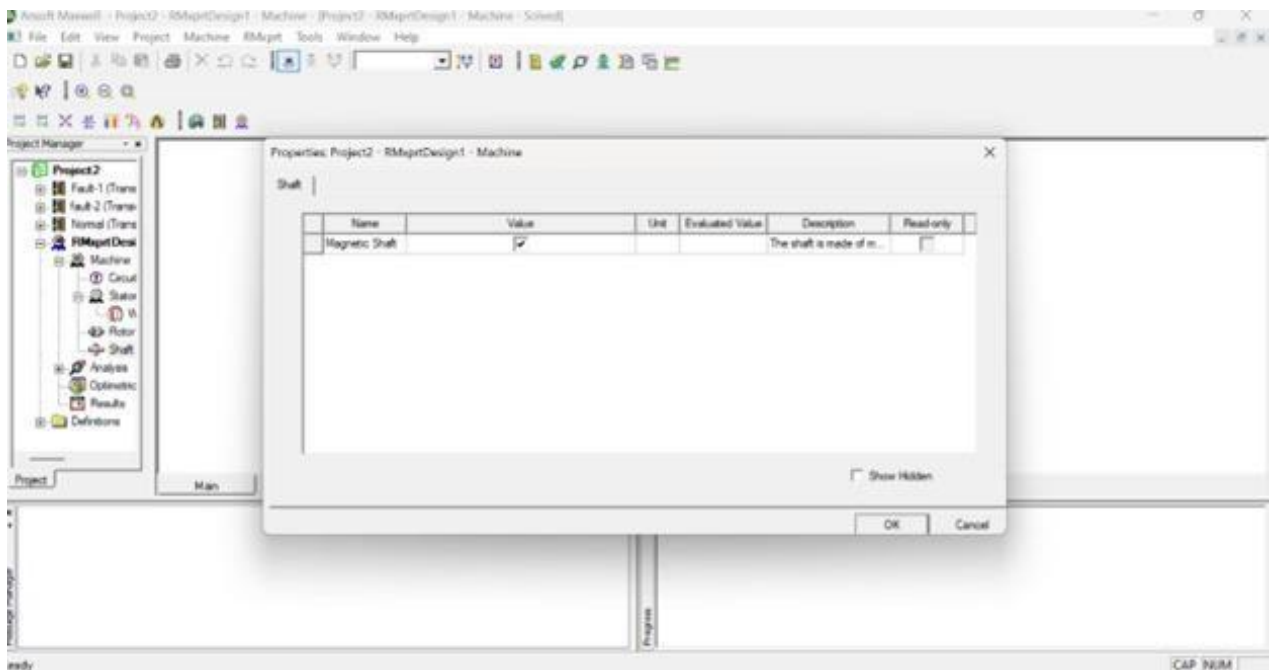


Fig.3.9 Shaft Parameters of SRM

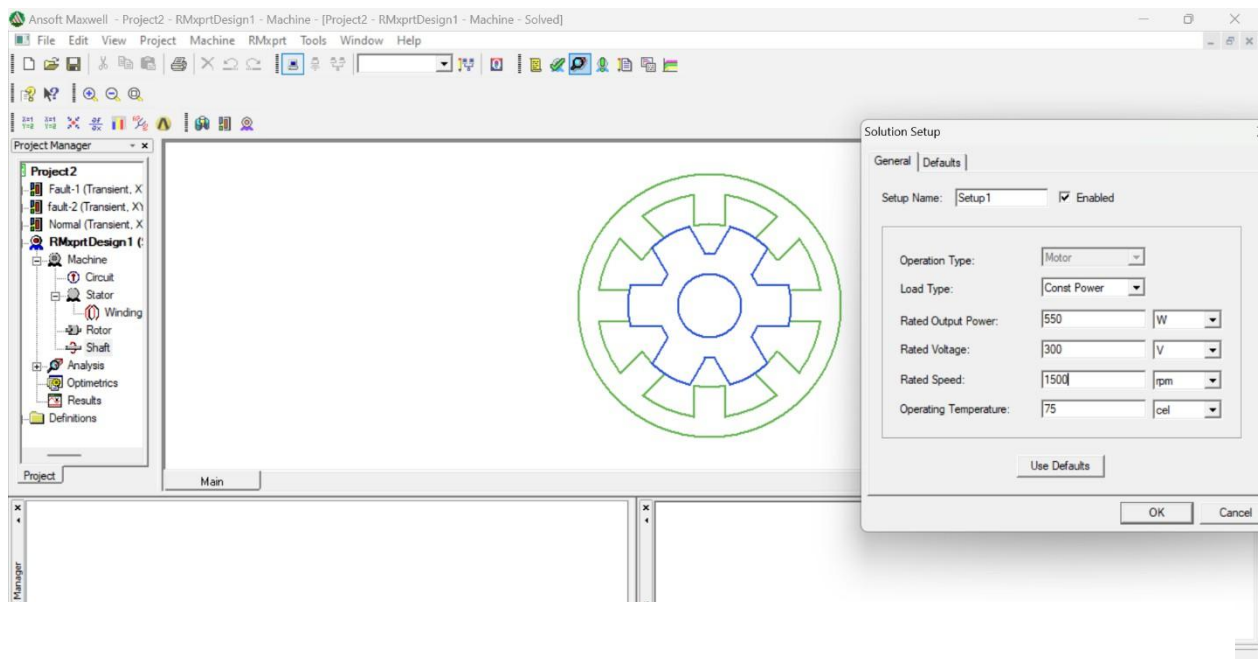


Fig.3.10 Setup configuration

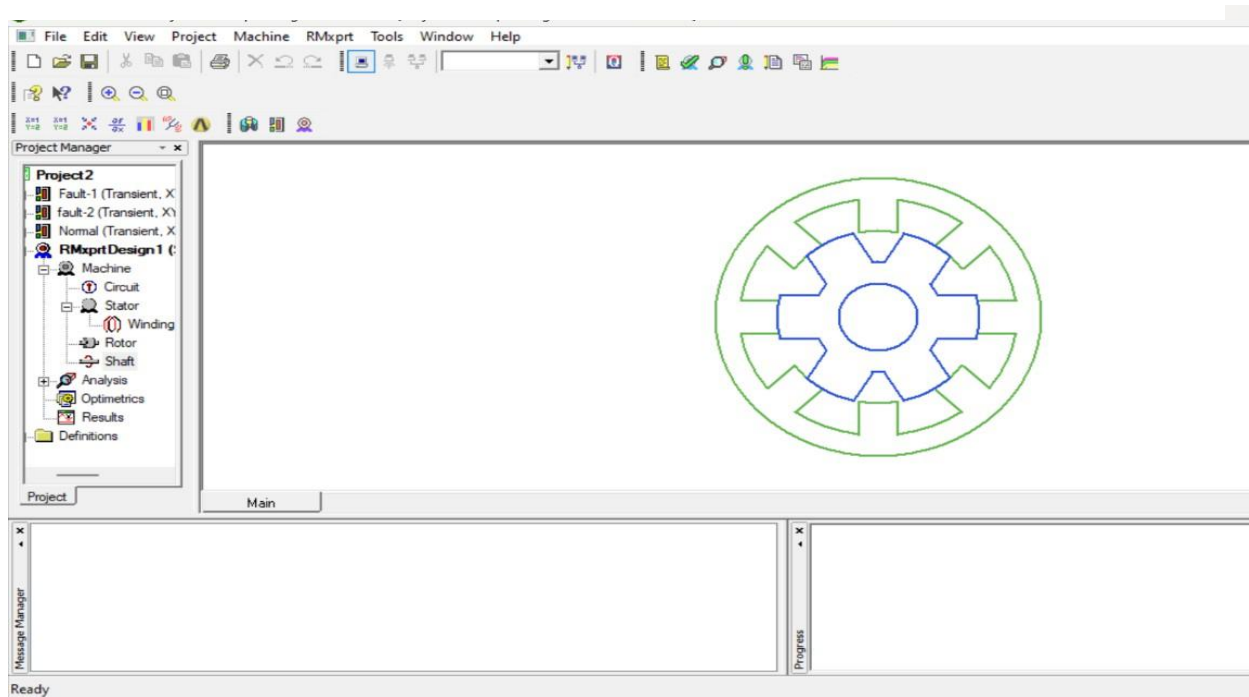


Fig.3.11 Designed Model of SRM

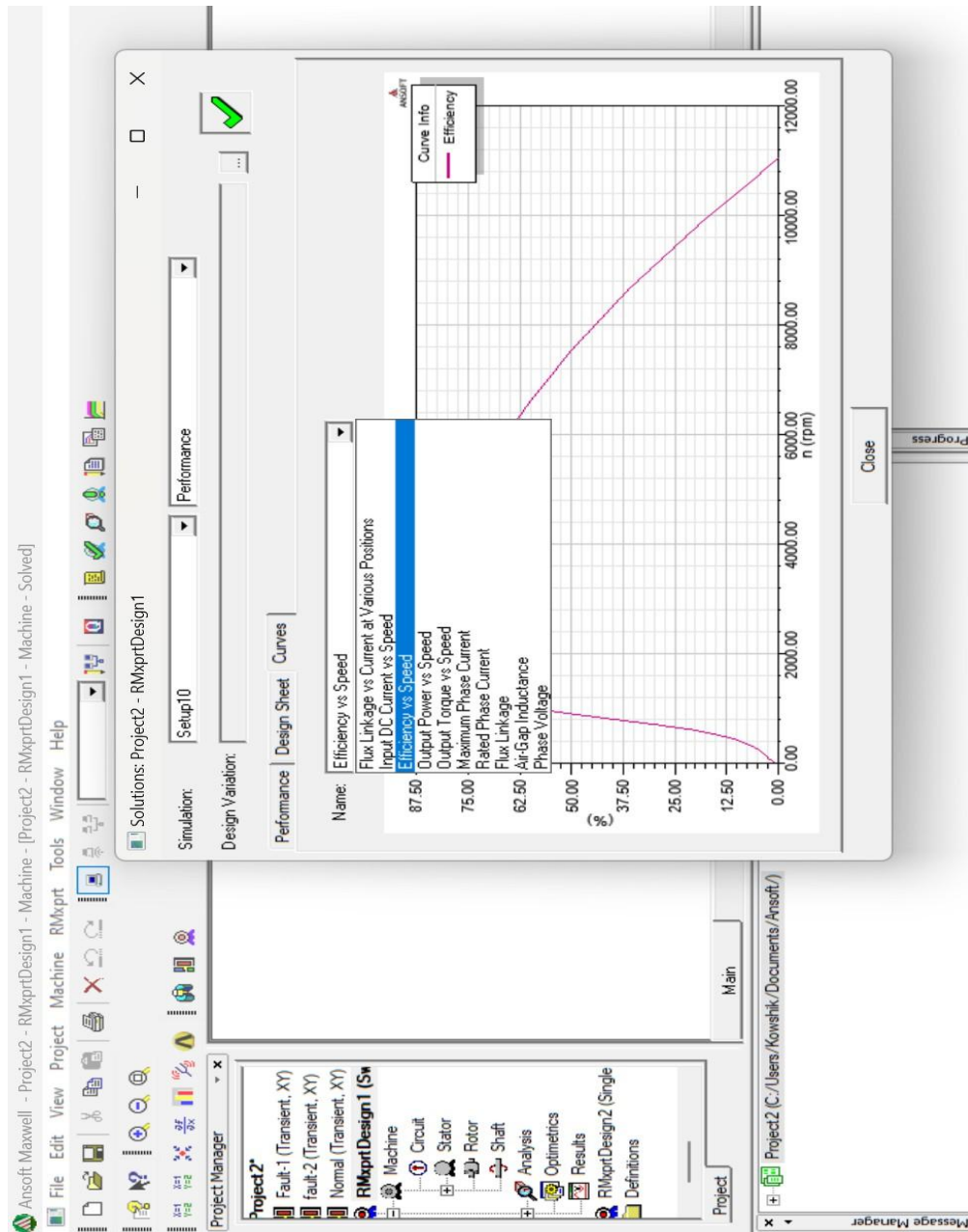


Fig.3.12 Performance of the SRM

After initialization of all the parameters run the simulation by adding setup then click on Graphs for the required performance of the motor. There are many Plotted analyses of the motor choose the required graph as shown in Fig.3.12. The output analysis of motor is simulation of given parameters, so change in parameters brings change in the output performance

3.2 2D Design of SRM

A 2D design of a Switched Reluctance Motor (SRM) is commonly used for initial analysis and optimization before moving to complex 3D simulations. In Maxwell Ansoft, a 2D model helps to study the electromagnetic behavior, torque characteristics, and flux distribution efficiently with reduced computational load. The design process involves defining the stator and rotor geometry, selecting the appropriate pole configuration, and setting material properties for the core and windings. The air gap, phase currents, and excitation pattern are optimized to improve performance. 2D simulations allow for accurate analysis of inductance profiles, magnetic saturation, and torque ripple.

Click on setup and choose create Maxwell Design and choose the setup that to be developed into 2D design as shown in Fig.3.13. Click on ok for further proceedings as shown in Fig and 2D design of the setup will be generated as shown in Fig.3.14.

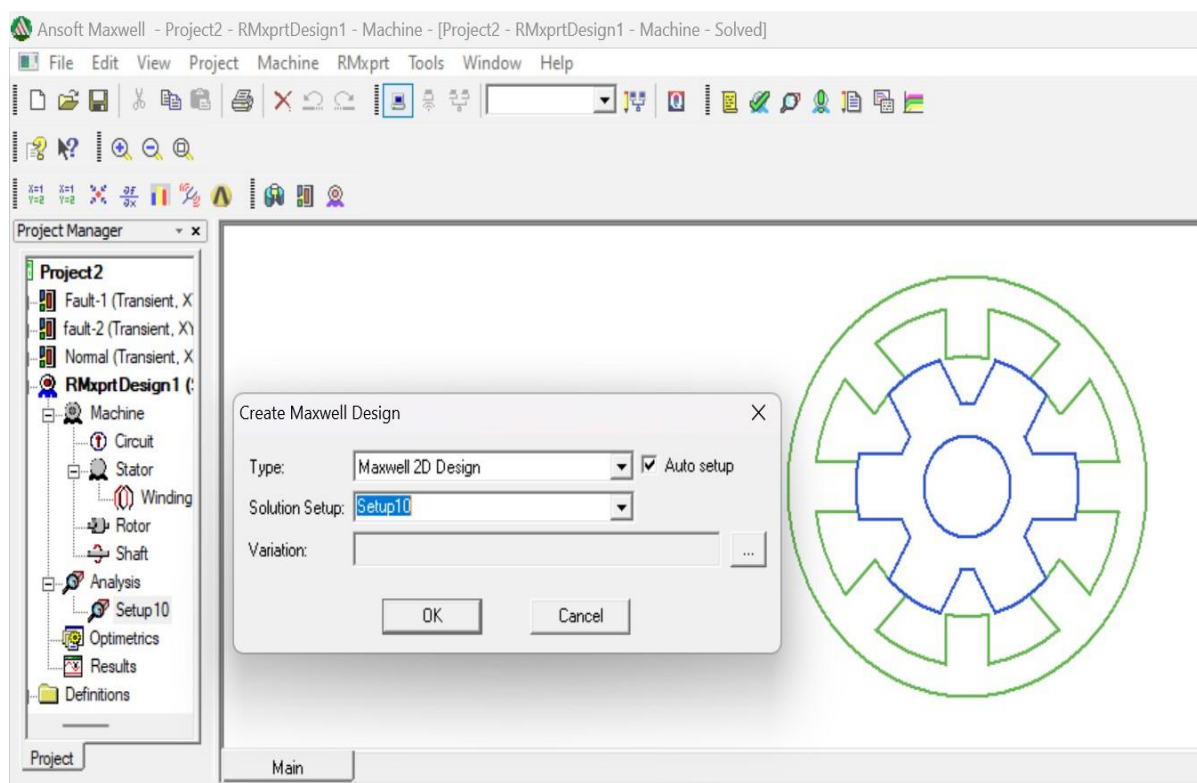


Fig.3.13 Creating Maxwell 2D Design

3.3 Fault generation in SRM

The fault condition in motor is generated by varying scale of rotor in SRM. After creation of 2D design select the rotor and click on edit then on scale as shown in Fig.3.15 which shows the drop box. The values are changed accordingly which changes the rotor dimensions and create fault conditions in SRM as shown in Fig.3.16 In order to obtain Torque and Windings current run the simulation by clearing errors and warnings. Follow the same process in-order to create multiple faults.

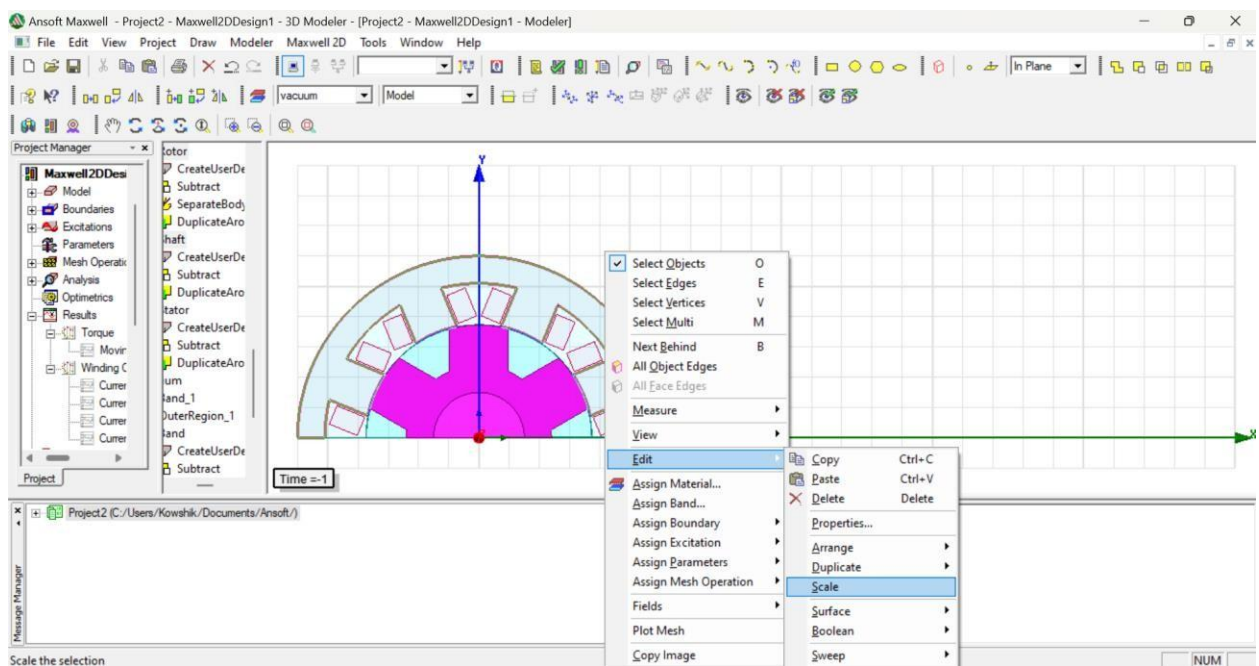


Fig.3.15 Selecting Rotor in SRM

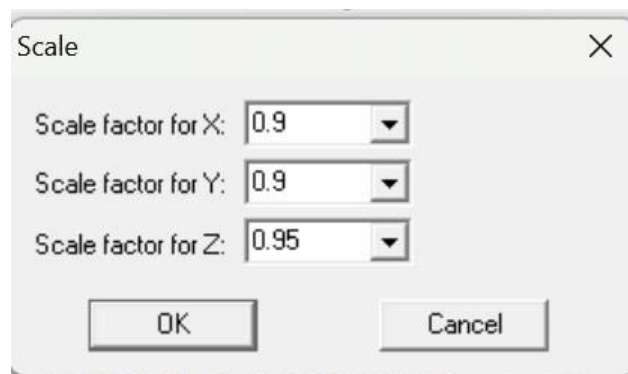


Fig.3.16 scale of the Rotor

CHAPTER 4

MACHINE LEARNING TECHNIQUES

4.1 Machine Learning Approach

Machine learning is used in this project to predict faults in a Switched Reluctance Motor (SRM) by analyzing torque and winding currents. These two parameters provide important information about the motor's performance. By studying how they change during normal operation and under fault conditions, our model can accurately detect when a fault occurs. This makes the fault detection process faster, more efficient, and reliable compared to traditional methods.

To improve the accuracy of fault detection, we use a data-driven approach instead of traditional methods that rely on fixed rules or predefined thresholds. Machine learning allows the model to learn from past data and recognize patterns, helping it adapt to different operating conditions. This enables real-time fault classification, which means the system can quickly detect faults as soon as they occur. Several machine learning algorithms can be utilized for fault detection in this project. Each algorithm has distinct strengths and applications, making them suitable for different aspects of fault classification. The following are some of the most commonly used machine learning algorithms for fault detection:

4.1.1 Linear Regression

Linear regression is one of the simplest machine learning algorithms, primarily used for predicting continuous values. It establishes a relationship between an independent variable (input) and a dependent variable (output) using a straight-line equation: $Y = mX + c$

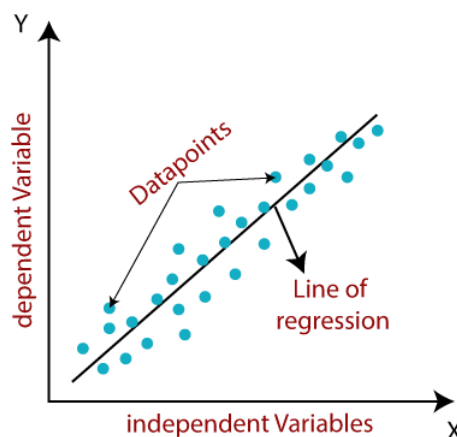


Fig.4.1 Linear Regression

In this project, linear regression can serve as a baseline model to analyze trends in torque and winding currents as shown in Fig.4.1. However, since fault classification involves discrete categories rather than continuous predictions, this model may not be the most effective choice for final implementation. Nevertheless, it can provide valuable insights into the overall behavior of the system.

4.1.2 Random Forest

The Random Forest algorithm is an ensemble learning method that constructs multiple decision trees and combines their outputs to enhance accuracy and stability as shown in Fig.4.2. This approach is particularly effective for classification tasks, making it a suitable choice for the SRM fault detection system. The algorithm functions by training multiple decision trees on different subsets of the dataset. When a new data point is introduced, each tree provides a prediction, and the final classification is determined through a majority vote. This process helps reduce overfitting, improve generalization, and make the model more resilient to variations in data.

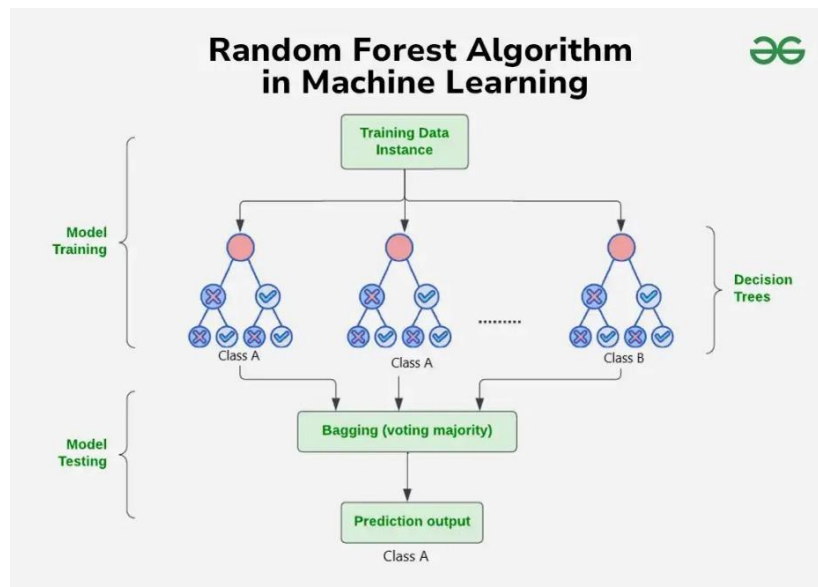


Fig.4.2 Random Forest

In this project, Random Forest is used to analyze torque and winding current values and classify them into normal or faulty conditions. By training the model on labelled data representing both normal and faulty operations, it learns to recognize fault signatures and make accurate predictions. Due to its ability to handle complex datasets and provide reliable fault classification, Random Forest has been chosen as the primary algorithm for this fault detection system.

4.1.3 Support Vector Machine (SVM)

SVM is a powerful classification algorithm that identifies the optimal hyperplane to separate different classes within a dataset. It is particularly useful when distinct patterns exist between normal and faulty conditions. One key advantage of SVM is its capability to work with both linear and nonlinear data. By utilizing kernel functions, it can transform data into higher dimensions to enhance separation between classes. In this fault detection project, SVM can assist in identifying specific patterns in torque and current variations, helping to distinguish between different fault conditions.

4.1.4 Artificial Neural Networks (ANN)

ANNs are deep learning models inspired by the structure of the human brain. They consist of multiple interconnected layers of neurons that process input data to recognize complex patterns. In fault detection applications, ANNs can be trained to recognize intricate fault signatures based on torque and winding current variations. As more data becomes available, the network improves its ability to distinguish between normal and faulty conditions. However, training neural networks requires significant computational resources and large amounts of labelled data.

4.1.5 Selection of the Best Algorithm for The Project

Each of these machine learning algorithms has unique strengths and limitations. For this Switched Reluctance Motor (SRM) fault detection system, Random Forest has been chosen as the primary model due to its:

- High classification accuracy and ability to handle complex datasets.
- Resilience to overfitting, which ensures reliable predictions.
- Ability to process large amounts of fault data efficiently.
- Ease of interpretation and implementation in real-time fault detection applications.

By integrating Random Forest into this fault detection system, the project aims to develop a robust and efficient real-time classification model that improves predictive maintenance, minimizes motor downtime, and enhances overall system reliability.

4.2 Algorithm Implementation

The first step in our process is data collection. We gather data from the SRM during normal operation as well as under four different fault conditions. This data includes values of torque and winding currents, which change when a fault is present. After collecting the data, we merge and preprocess it to ensure accuracy and remove any unnecessary or incorrect data points. This step is important because clean and well-organized data helps the machine learning model perform better.

Then, we use the Random Forest algorithm to train our model for fault detection. Random Forest is an ensemble learning method that creates multiple decision trees and combines their results to improve accuracy and reduce errors.

Each decision tree in the Random Forest learns from a different subset of data. When new data is given to the model, each tree makes a prediction, and the final result is determined by a majority vote (for classification) or an average (for regression). This method makes Random Forest highly accurate and resistant to overfitting, meaning it performs well even on complex datasets.

In our project, Random Forest analyses torque and winding current values to identify patterns associated with different fault conditions. By training on both normal and faulty data, the algorithm learns to classify new data correctly, helping us detect faults in the Switched Reluctance Motor (SRM) with high reliability. This makes it an excellent choice for our fault detection system, ensuring quick and accurate fault classification.

Once the model is trained, it can analyse new data and determine whether the SRM is operating normally or experiencing a fault. If a fault is detected, the system can alert users, allowing them to take action before the motor is damaged further.

By using machine learning in our fault detection system, we aim to improve motor reliability, reduce downtime, and enhance predictive maintenance. This approach ensures that faults are detected early, helping to prevent failures and making the system more efficient.

4.2.1 Python for Machine Learning

Python has become one of the most widely used programming languages for machine learning and artificial intelligence due to its simplicity, versatility, and vast collection of libraries. In this project, Python is being used to develop a machine learning model that detects faults in a Switched Reluctance Motor (SRM) by analysing torque and winding current data. By leveraging Python's powerful tools, the model can learn from historical data and make accurate predictions about the motor's condition.

One of the key reasons for choosing Python is its user-friendly syntax, which makes it easy to write, understand, and modify code. Unlike other programming languages that require complex implementations, Python provides built-in functions and pre-written libraries that simplify the entire machine-learning workflow. This allows for faster development and testing, ensuring that the model is both efficient and accurate.

In this project, Python plays a crucial role at different stages of model development. First, it is used for data collection and preprocessing, where libraries like NumPy and Pandas help organize and clean the data. Since machine learning models perform best when trained on high-quality data, this step ensures that the input features, such as torque and winding currents, are structured properly before training.

Next, Python is used for training the model using the Random Forest algorithm, which is available in the Scikit-learn library. Random Forest is chosen because of its ability to handle complex data, reduce overfitting, and provide reliable predictions. The model learns fault patterns by analysing past data and then applies this knowledge to classify new data as either normal or faulty.

After training, the model's performance is evaluated and optimized using Python's built-in functions for accuracy measurement. If necessary, adjustments are made to improve prediction accuracy. Visualization also plays a crucial role in understanding how the model is performing. With the help of Matplotlib and Seaborn, various graphs and charts are generated to analyse the model's output, making it easier to identify any errors or inconsistencies.

By using Python, the project benefits from a flexible, efficient, and well-supported development environment. It allows seamless integration of different machine learning techniques, quick experimentation with different models, and the ability to handle large datasets efficiently. This ensures that the fault detection model for the Switched Reluctance

Motor is not only accurate but also robust and scalable. Ultimately, this approach helps in reducing unexpected failures, improving motor efficiency, and ensuring a more reliable predictive maintenance system.

4.2.2 Jupyter Notebook for model development

For this project, Jupyter Notebook is being used to develop a machine learning model that can predict faults in a Switched Reluctance Motor (SRM) based on torque and winding current data.

Jupyter Notebook is an interactive coding environment that allows users to write and execute code in small sections called cells. This feature makes it easier to test different approaches, make changes, and visualize results in real-time. Instead of writing and running the entire program at once, Jupyter Notebook enables step-by-step execution, helping to analyse and debug the code efficiently. This is especially useful in machine learning, where working with large datasets and tuning model parameters requires constant experimentation.

In this project, the development of the SRM fault detection model follows a structured approach:

Data Collection and Preprocessing: The first step involves gathering data from the SRM under both normal and faulty conditions. This data is then cleaned and pre-processed to remove inconsistencies, ensuring that the machine learning model learns from accurate and meaningful information.

Feature Selection: The most relevant parameters, such as torque and winding currents, are chosen as input features. These features provide crucial insights into the motor's health, helping the model distinguish between normal and faulty conditions.

Model Training: A supervised learning algorithm, Random Forest, is used to train the model. This algorithm learns from labeled data, recognizing patterns that indicate different fault conditions. Since Random Forest is known for its high accuracy and ability to handle complex datasets, it is well-suited for this task.

Evaluation and Optimization: After training, the model's performance is tested using various evaluation metrics. This step ensures that the model can accurately classify new data and detect faults in real-world scenarios. If needed, optimizations are made to improve accuracy and reliability.

Visualization and Interpretation: One of the major advantages of using Jupyter Notebook is its ability to generate visual outputs, such as graphs and charts. These visualizations help in understanding how the model is performing, making it easier to interpret results and make necessary improvements.

By using Jupyter Notebook, this project benefits from a flexible, interactive, and user-friendly environment that simplifies the entire machine learning workflow. It allows for better documentation, easy debugging, and efficient experimentation, ultimately leading to a robust and reliable fault detection model for the Switched Reluctance Motor. This approach not only enhances predictive maintenance but also helps in minimizing unexpected failures and improving the overall efficiency of the motor system.

4.2.3 Libraries used

Developing a machine learning model for fault detection in a Switched Reluctance Motor (SRM) requires the use of powerful tools that can handle large amounts of data, train models effectively, and make accurate predictions. Since the goal of this project is to classify whether the motor is operating normally or experiencing a fault, various Python libraries have been incorporated to streamline the entire process. Each library has a specific role, from loading and processing data to training the machine learning model and evaluating its performance.

By using a combination of Pandas, NumPy, and Scikit-learn, the project leverages well-established machine learning techniques to detect motor faults based on torque and winding currents. The use of the Random Forest algorithm ensures high accuracy and robustness in predictions, making this approach much more effective than traditional threshold-based fault detection methods. Below is a detailed breakdown of the libraries used and their significance in this project.

1. Pandas

One of the most critical steps in machine learning is data preprocessing, where raw data is structured and prepared before being used to train the model. Pandas is an essential library in Python that simplifies this process by providing a Data Frame structure, which allows for easy organization, filtering, and modification of data. In this project, Pandas is used to load and clean the dataset before feeding it into the machine learning model.

The dataset, which is stored in a CSV file, is read into a Data Frame using Pandas functions. Since some column names in the dataset might be long or difficult to read, Pandas allows renaming them for better readability.

Additionally, fault conditions in the dataset are converted into numerical values so they can be used in machine learning models. If the motor is operating under faulty conditions, it is assigned a label of 1 (faulty); otherwise, it is labeled 0 (normal). This transformation is crucial because machine learning algorithms require numerical inputs rather than text labels.

2. NumPy

Since machine learning models involve a large number of mathematical operations, it is essential to use a library that can perform calculations quickly and efficiently. NumPy is a fundamental library that provides support for multi-dimensional arrays and mathematical functions, making numerical operations much faster. In this project, NumPy is mainly used to format the input data for making predictions. When the user provides real-time torque and winding current values, these values are converted into a NumPy array before being passed to the trained machine learning model.

3. Scikit-learn

Scikit-learn (or sklearn) is one of the most widely used libraries for machine learning in Python. It provides a variety of tools for data preprocessing, model training, and evaluation, making it ideal for building predictive models. By keeping 20% of the dataset for testing, we ensure that the model is not just memorizing the data but actually learning patterns that will help it generalize well to new cases.

4. Random Forest Classifier

For this project, a model is needed that can accurately classify faults in an SRM based on torque and winding current values. Among various machine learning algorithms, Random Forest was chosen because of its high accuracy and robustness.

5. Evaluating the Model with Accuracy Score

Once the model is trained, it is essential to check how well it performs. The accuracy score function calculates the percentage of correct predictions made by the model. This metric gives insight into how effectively the model can classify fault conditions in the SRM. A high accuracy score means the model is reliable, while a low score suggests the need for further improvements.

CHAPTER 5

ML MODEL FOR FAULT ANALYSIS IN SRM

By implementing a Random Forest Classifier, the system learns from historical data to distinguish between normal and faulty conditions. The goal is to enhance fault detection accuracy, minimize downtime, and improve overall motor reliability.

This section provides a detailed explanation of the Python code used to develop the model, including data preprocessing, model training, evaluation, and real-time fault prediction.

5.1 Importing Required Libraries

The first step in the code is importing the necessary Python libraries, which play essential roles in data handling, machine learning model development, and evaluation.

pandas: This library is used to handle structured data in the form of tables (DataFrames). It allows for efficient reading, manipulation, and processing of datasets stored in CSV files.

Numpy : This numerical computing library helps in handling arrays and performing mathematical operations efficiently. It is used for structuring input data for the machine learning model.

train_test_split (from scikit-learn): This function is used to split the dataset into training and testing sets, ensuring that the model is properly trained and evaluated.

RandomForestClassifier (from scikit-learn): This machine learning algorithm is used to classify motor conditions. It consists of multiple decision trees, making it robust and accurate.

accuracy_score (from scikit-learn): This function evaluates the performance of the model by calculating how often the model's predictions match the actual data labels.

These libraries work together to enable data processing, machine learning model training, and performance evaluation, making them essential for this project.

5.2 Loading and Preprocessing the Dataset

The dataset used in this project is stored in a CSV (Comma-Separated Values) file, which contains information about the motor's operating conditions under different scenarios. The `load_dataset(file_path)` function is responsible for:

- Reading the CSV file using pandas and storing it in a DataFrame.
- Renaming columns to make them more understandable (e.g., "Time [ms]" is renamed to "time", and "Moving1.Torque [NewtonMeter]" is renamed to "torque").
- Converting time from milliseconds to seconds, if necessary, to maintain consistency in units.
- Creating a new column called "fault_label", which assigns numerical values to motor conditions:
 - A value of 1 represents a faulty condition.
 - A value of 0 represents a normal condition.

This preprocessing step is essential because machine learning models work with numerical data, and categorical values (such as "normal" and "fault") need to be converted into numerical labels for effective learning.

5.3 Defining Features and Target Variable

After loading and preprocessing the dataset, the next step is selecting features (input variables) and defining the target variable (output).

Features (Inputs): These are the parameters that influence fault detection. The selected features include:

Torque: The rotational force applied to the motor.

Current in Phase A, B, C, and D: The current flowing through each phase of the motor, which changes under different operating conditions.

Target Variable (Output): The column `fault_label` is used as the target variable, indicating whether the motor is functioning normally or experiencing a fault.

By defining these inputs and outputs, the model understands what data to analyze and what outcome to predict.

5.4 Splitting the Dataset into Training and Testing Sets

To develop a reliable machine learning model, the dataset is divided into two parts:

Training Set (80%): This portion of the dataset is used to teach the model how to recognize fault patterns based on the provided features.

Testing Set (20%): This portion is used to evaluate the model's performance on unseen data, ensuring that it generalizes well.

The `train_test_split()` function ensures a random split, preventing any bias in the model's learning process. A random seed (`random_state=42`) is also set to ensure reproducibility, meaning that every time the code is run, the dataset is split in the same way.

5.5 Training the Machine Learning Model

The Random Forest Classifier is used as the machine learning algorithm in this project. It is a popular classification algorithm because it:

- Consists of multiple decision trees, which improves accuracy and reduces errors.
- Is resistant to overfitting, meaning it performs well on both training and unseen data.
- Can handle complex relationships between features, making it ideal for real-world fault detection.

The model is initialized with 100 decision trees, and then it is trained using the `fit()` function with the training dataset (`X_train, y_train`). During training, the model learns to identify fault patterns in the data.

5.6 Evaluating the Model's Accuracy

After training, the model's accuracy is tested using the testing dataset. The trained model makes predictions on `X_test`, and these predictions are compared to the actual fault labels (`y_test`) using the `accuracy_score()` function.

The accuracy score is then printed as a percentage, showing how well the model classifies normal and faulty conditions. A high accuracy score indicates that the model is performing well, whereas a low score suggests that improvements (e.g., more data or better feature selection) may be needed.

5.7 Creating a Function for Real-Time Fault Prediction

A function named `predict_fault()` is created to allow users to manually enter values for torque and phase currents (A, B, C, and D) to check the motor's condition.

- The user enters numerical values for torque and current in each phase.
- The input values are stored in an array and formatted so that the model can process them correctly.
- The trained model predicts the fault condition based on the provided input.
- The output is displayed to the user, showing whether the motor is in a normal state or experiencing a fault.

Additionally, the function includes error handling to prevent crashes if the user enters invalid data. If an error occurs, a message is displayed to inform the user about the issue.

5.8 Running the Fault Prediction Function

Finally, the program calls the `predict_fault()` function, allowing users to enter real-time data and receive a prediction. This step ensures that the model is not just a theoretical tool but a practical solution for real-world fault detection. This interactive approach makes the system useful for engineers and maintenance teams, allowing them to quickly diagnose motor conditions and take necessary actions

CHAPTER 6

RESULTS & DISCUSSION

6.1 Simulation results

The Design of the motor is simulated and the performance of the Motor is observed by various graph analysis like Efficiency Graph and Flux Linkage vs current at various positions as shown in Fig.6.1 and Fig.6.2 This provides better understanding about the motor and knowledge on the characteristics and behavior of the motor

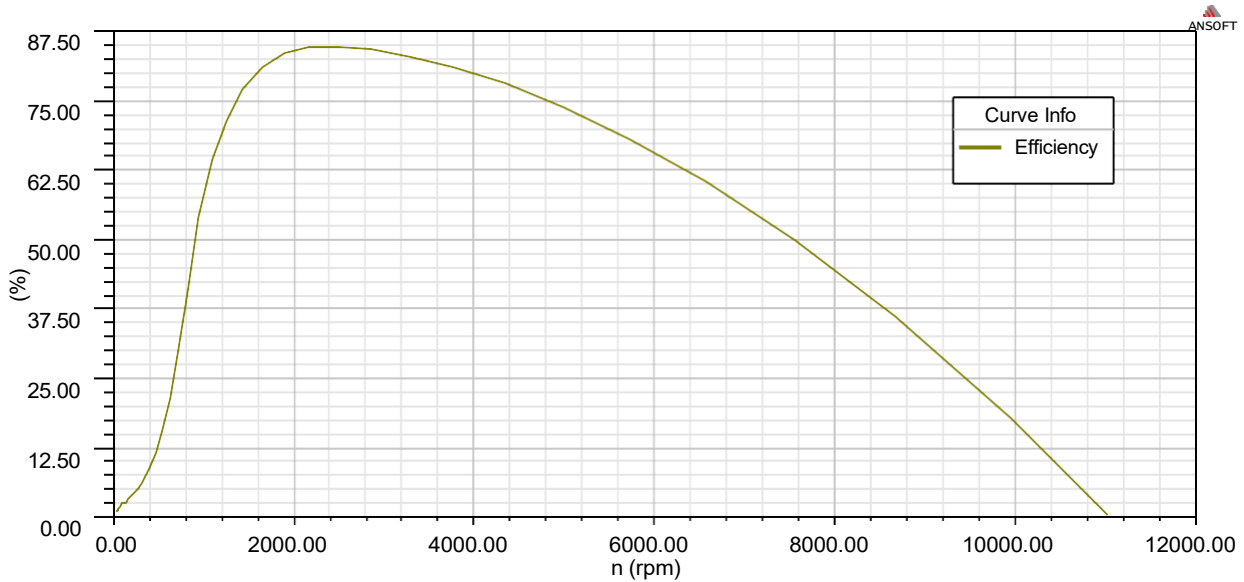


Fig.6.1 Efficiency graph of SRM

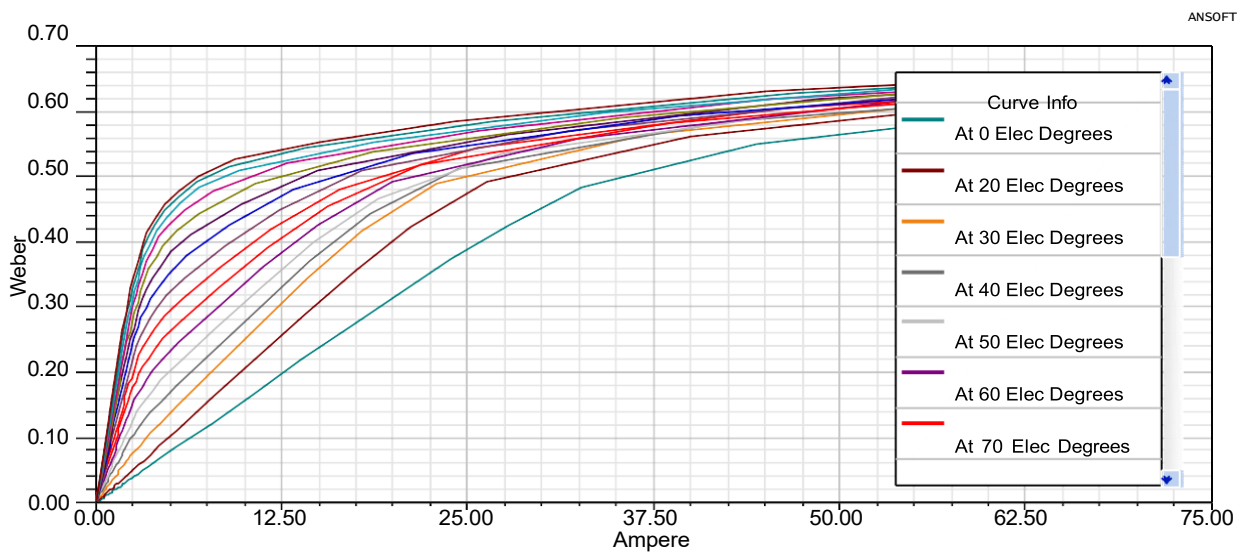


Fig.6.2 Flux linkage vs current at various positions

The motor is examined at normal and faulty conditions. The graphs show the Torque and Winding currents of its respective condition. Fig.6.3 and Fig.6.4 displays the Graph of the Motor at the normal condition.

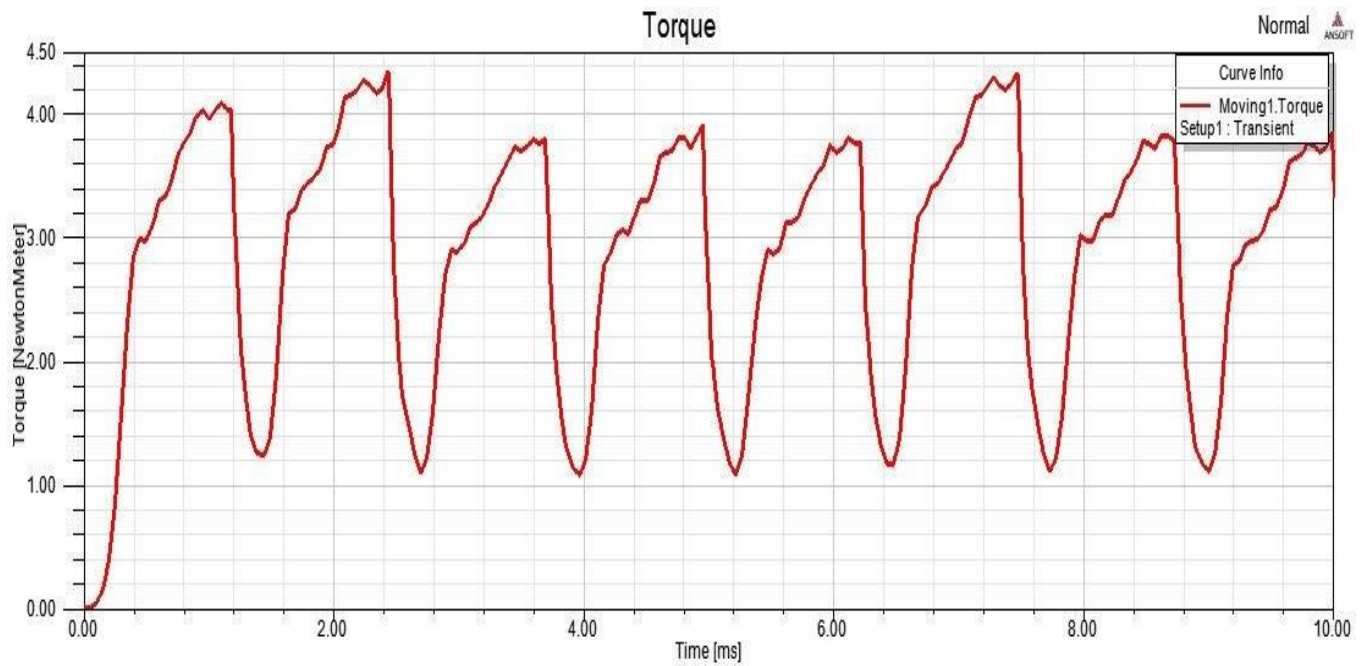


Fig.6.3. Torque graph at normal condition

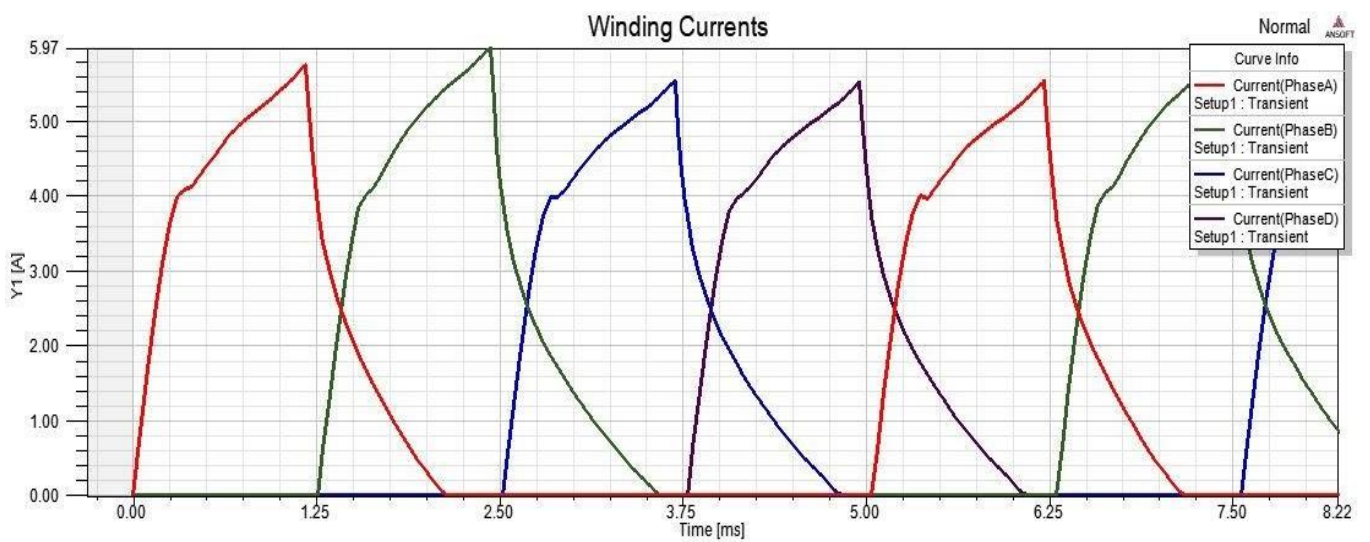


Fig.6.4. Windings current at normal condition

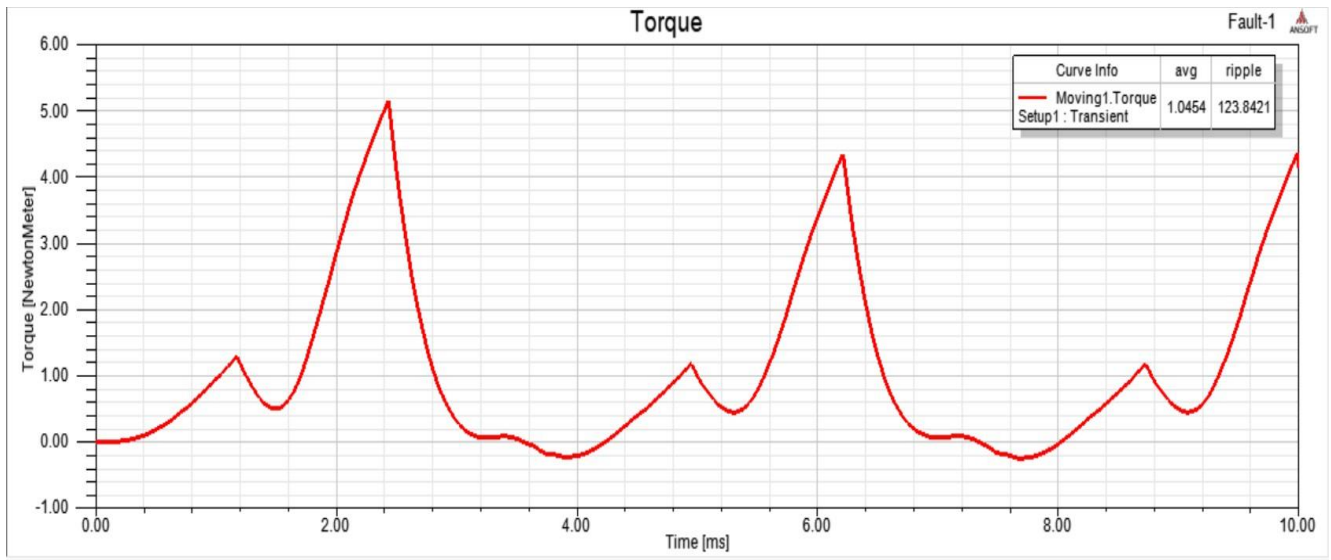


Fig.6.5 Torque at Fault-1 condition

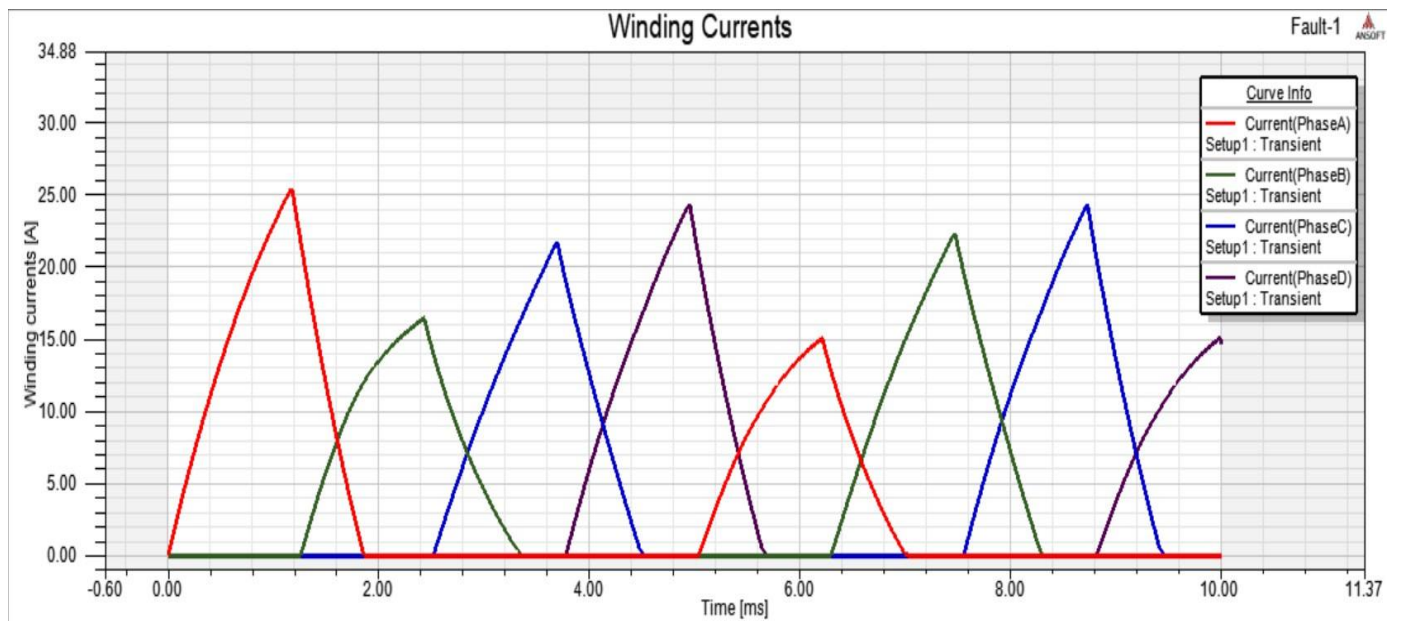


Fig.6.6 Windings current at Fault-1 condition

The graphs show the analysis of torque and winding currents at Fault-1 condition as shown Fig.6.5 and Fig.6.6 The 4 different Colors in the Windings current graph give information about the 4-phase current. The data is Retrieved from the graph as a data sheet.

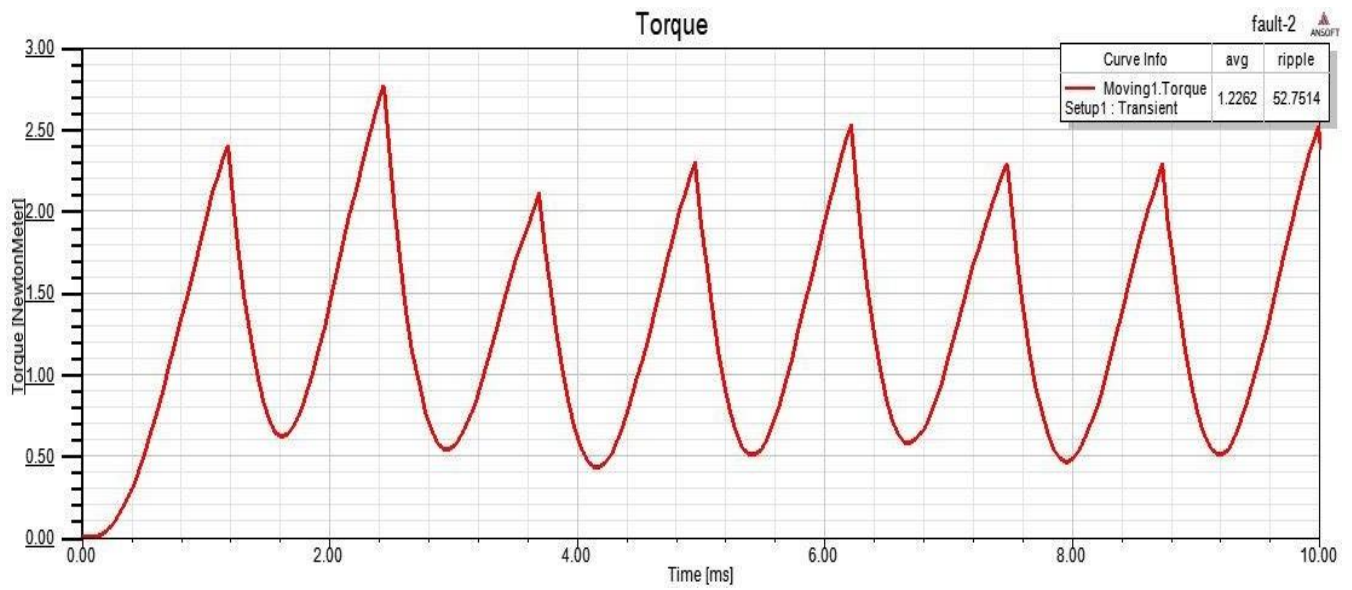


Fig.6.7 Torque at Fault-2 condition

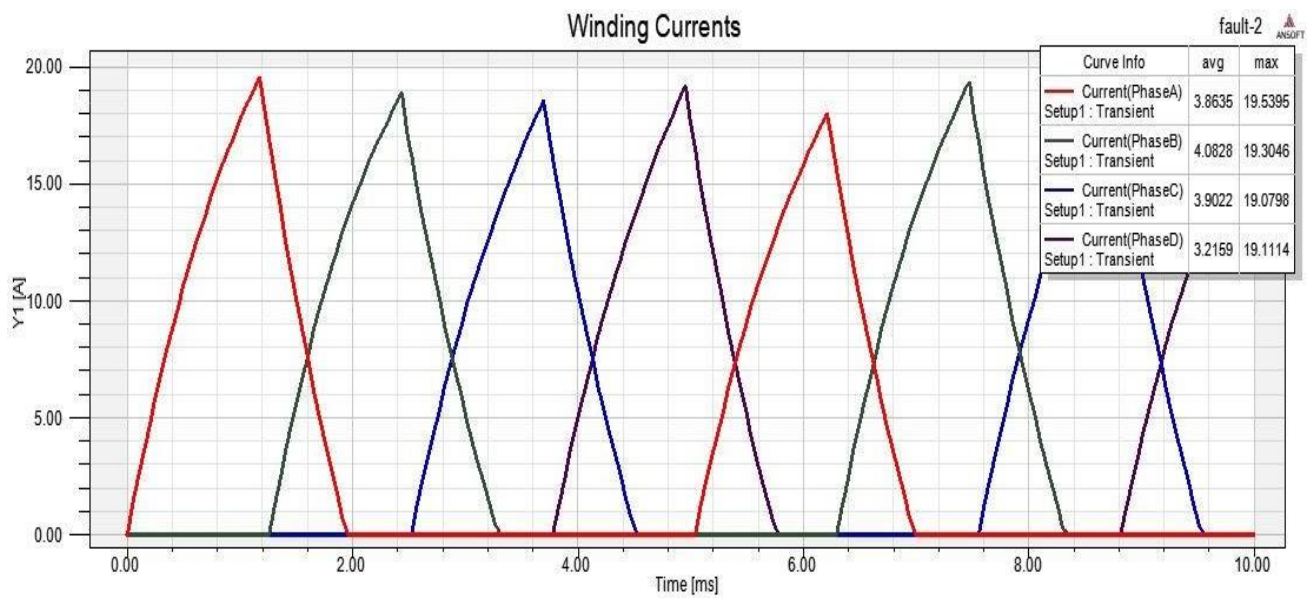


Fig.6.8 Windings current at Fault-2 condition

The graphs show the analysis of torque and winding currents at a different fault condition as shown Fig.6.7 and Fig.6.8. The graphs are clearly not identical and are deviated from normal condition graphs.

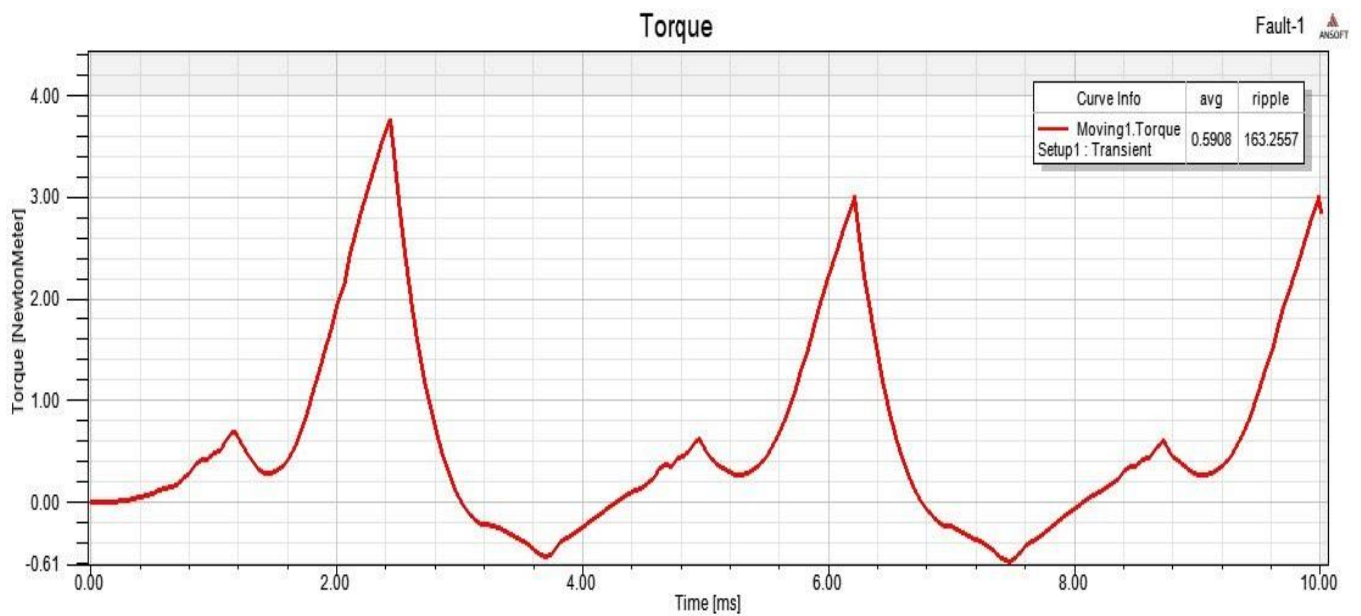


Fig.6.9 Torque at Falt-3 condition

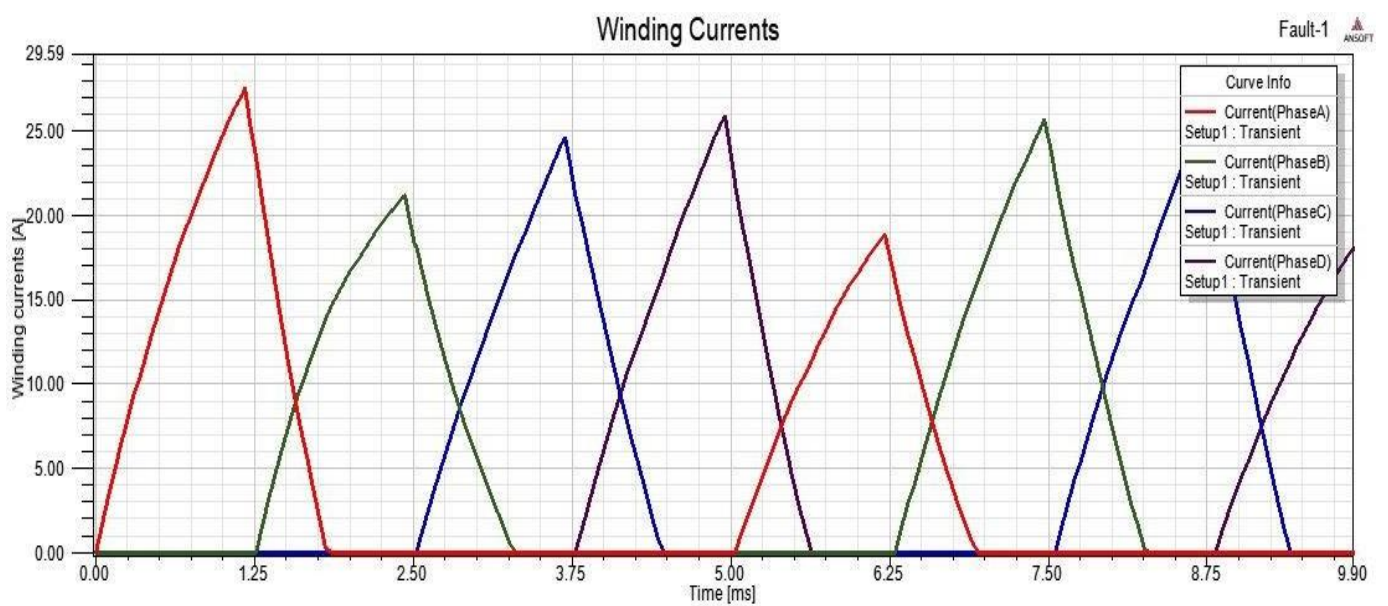


Fig.6.10 Windings current at Fault-3 condition

The graphs show the analysis of torque and winding currents at Fault-3 condition as shown Fig.6.9 and Fig.6.10. These graphs are obtained by running the motor at another fault condition.

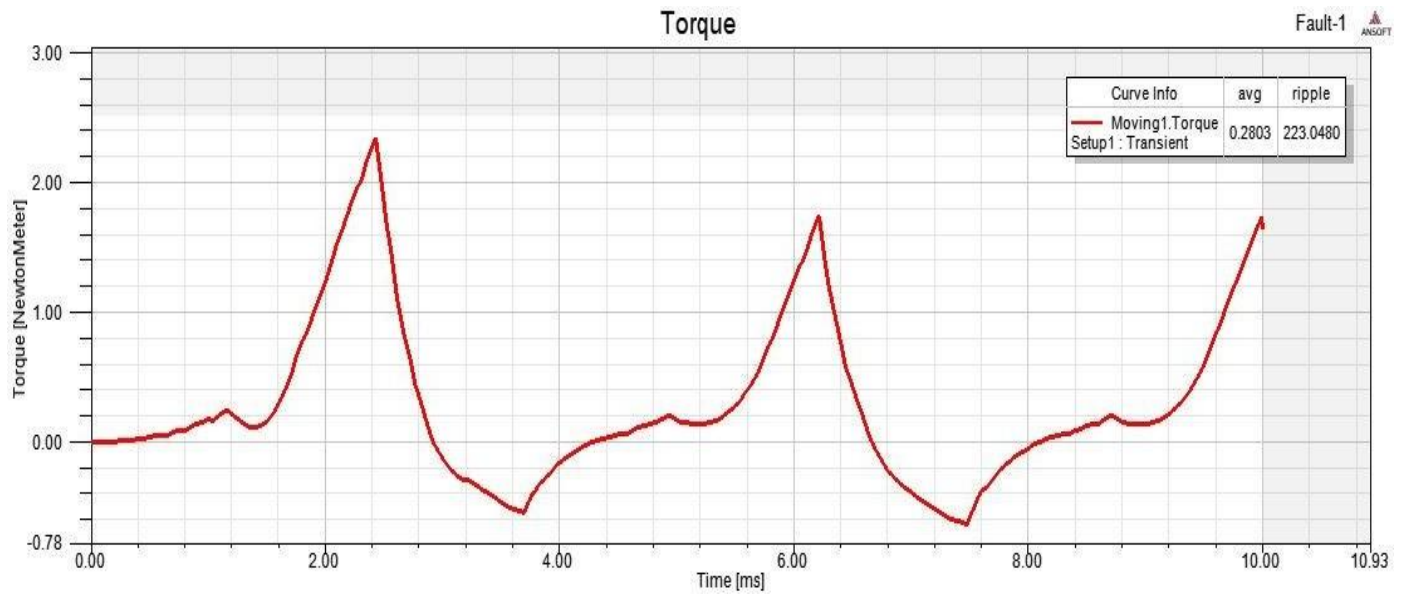


Fig.6.11 Torque at Fault-4 condition

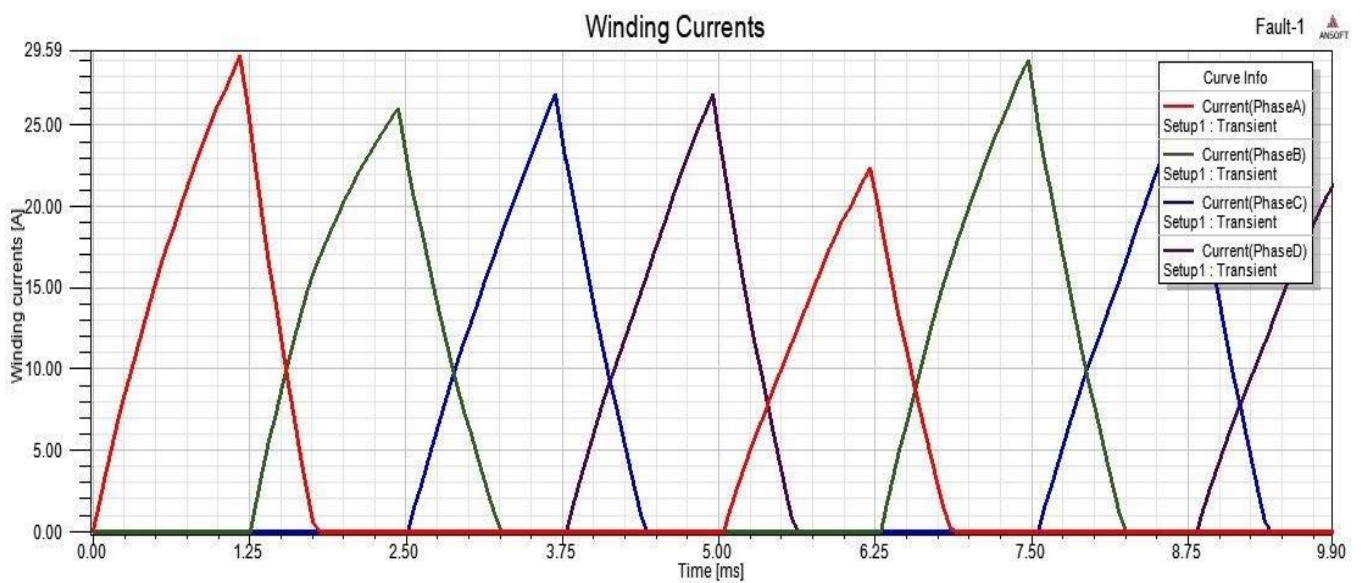


Fig.6.12 Windings current at Fault-4 condition

The graphs show the analysis of torque and winding currents at Fault-4 condition as shown Fig.6.11 and Fig.6.12. These graphs are obtained by running the motor at fault condition similar to Fault-1,2 and 3, to obtain more fault condition current and torque values.

6.2 Machine learning model results

The Algorithm machine learning model is developed. Data sets are given to the ML which examines the condition of torques and windings currents at normal and different conditions of faults. The code is developed in such a way that it not only shares the information about the fault or normal condition of the motor but also provide the information about where the fault is present like in which phase. It also provides the severity level of fault generated in the motor as the levels are divided into 4 stages based on the fault deviation. The 4 stages are low, Medium, High, severally High. The model contains accuracy determination that shows the possibility of the state to be high. Different faults are subjected to evaluation and the algorithm results are depicted in Fig. 6.13 - 6.17.

```
# Identify phase with max deviation
faulty_phase = max(deviations, key=deviations.get)
deviation_value = deviations[faulty_phase]
severity = get_fault_severity(deviation_value)

print(f"⚠️ Fault likely in Phase {faulty_phase}")
print(f"🔴 Fault Severity: {severity}")
else:
    print("\n✅ SRM Condition: NORMAL")

except Exception as e:
    print(f"❌ Error in prediction: {e}")

predict_fault()
```

Model Accuracy: 99.18%

Enter Moving Torque value: 4.010383642
Enter Phase A Current: 5.733165806
Enter Phase B Current: 0.000153338
Enter Phase C Current: 0.000152544
Enter Phase D Current: 0.00013313

✅ SRM Condition: NORMAL

Fig. 6.13 Machine learning output at normal state

The machine learning model is validated by giving normal state inputs obtained by the dataset by running the motor at normal condition. The model prediction can be seen in Fig. 6.13.

```

        severity = get_fault_severity(deviation_value)

        print(f"⚠️ Fault likely in Phase {faulty_phase}")
        print(f"🔴 Fault Severity: {severity}")
    else:
        print("\n✅ SRM Condition: NORMAL")

    except Exception as e:
        print(f"❌ Error in prediction: {e}")

predict_fault()

```

Model Accuracy: 99.18%

Enter Moving Torque value: 1.286263759
Enter Phase A Current: 19.5233129
Enter Phase B Current: 0.000155617
Enter Phase C Current: 0.000148171
Enter Phase D Current: 0.000138465

C:\Users\ravir\Desktop\Lib\site-packages\sklearn\utils\validation.py:2739: UserWarning: X does not have valid feature names, but RandomForestClassifier was fitted with feature names
warnings.warn(

🔴 Fault Detected!
⚠️ Fault likely in Phase A
🔴 Fault Severity: Extremely Severe

Fig.6.14 Machine learning output at Fault in Phase A

The machine learning model is validated by giving normal state inputs obtained by the dataset by running the motor at faulty condition. The model prediction of the state of the motor as well as the phase of which the Fault is observed can be seen from Fig. 6.14 – 6.17.

```

        print(f"🔴 Fault Severity: {severity}")
    else:
        print("\n✅ SRM Condition: NORMAL")

    except Exception as e:
        print(f"❌ Error in prediction: {e}")

predict_fault()

```

Model Accuracy: 99.18%

Enter Moving Torque value: 3.845993758
Enter Phase A Current: 0.000142492
Enter Phase B Current: 16.06085671
Enter Phase C Current: 0.000150598
Enter Phase D Current: 0.000147997

🔴 Fault Detected!
⚠️ Fault likely in Phase B
🔴 Fault Severity: Severely High

Fig.6.15 Machine learning output at fault in phase B


```
else:
    print("\n✅ SRM Condition: NORMAL")

except Exception as e:
    print(f"❌ Error in prediction: {e}")

predict_fault()
```

Model Accuracy: 99.18%

Enter Moving Torque value: -0.160510975
Enter Phase A Current: 0.000153041
Enter Phase B Current: 0.0002453646
Enter Phase C Current: 14.09263
Enter Phase D Current: 0.00634345

- Fault Detected!
- ▲ Fault likely in Phase C
- Fault Severity: Severely High

Fig.6.16 Machine learning output at fault in phase C

Model Accuracy: 99.18%

Enter Moving Torque value: 2.9746554330
Enter Phase A Current: 0.0000734554
Enter Phase B Current: 0.06333344
Enter Phase C Current: 0.00000562443
Enter Phase D Current: 14.74545232

- Fault Detected!
- ▲ Fault likely in Phase D
- Fault Severity: Severely High

Fig.6.17 Machine learning output at fault in phase D

The model has been validated by the faults in different phases from already labelled data. The algorithm is sufficiently accurate and has predicted the faults in different phases without any errors.

CHAPTER 7

CONCLUSION & FUTURE SCOPE

In this project, a machine learning-based approach for fault detection in Switched Reluctance Motors (SRMs) has been successfully developed and analyzed. By utilizing Finite Element Method (FEM) simulations, key motor parameters such as current, flux, torque, and induced voltage were studied under various operating conditions to create an accurate and comprehensive dataset. The application of the Random Forest algorithm enabled efficient and reliable identification of motor faults, proving to be significantly faster and more effective than traditional manual monitoring methods. This approach contributes to reducing unplanned downtime, extending the operational lifespan of the motor, and improving overall energy efficiency. The proposed system lays the foundation for real-time dynamic fault detection, enabling timely interventions and promoting autonomous maintenance solutions. The successful implementation of this project highlights the potential of AI-driven monitoring systems in ensuring the stable, safe, and sustainable operation of SRMs in various industrial applications.

The proposed fault detection system using machine learning and Finite Element Method (FEM) offers significant potential for future development. One major scope is its real-time implementation in industrial applications, enabling continuous monitoring and immediate fault detection. Integrating the system with IoT platforms can facilitate remote diagnostics and predictive maintenance. The methodology can also be extended to other motor types, enhancing its versatility. Future improvements may include the use of deep learning algorithms for better accuracy and the estimation of fault severity. Additionally, a user-friendly interface and adaptive learning models could make the system more robust and accessible. Overall, this project lays the foundation for intelligent, efficient, and scalable motor health monitoring solutions.

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